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LIST OF ABBREVIATION

Abbreviation	Full Form
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
Groq	Groq Cloud AI Inference Platform
JSON	JavaScript Object Notation
LLM	Large Language Model
Llama	Large Language Model Meta AI
ML	Machine Learning
NLP	Natural Language Processing
OCR	Optical Character Recognition
PyMuPDF	Python Library for PDF Processing
RAG	Retrieval-Augmented Generation
RAM	Random Access Memory
ROUGE	Recall-Oriented Understudy for Gisting Evaluation

ABSTRACT

The rapid growth of scientific publications has made it increasingly difficult for researchers, students, and professionals to efficiently review and analyze large volumes of academic literature. Traditional research paper summarization tools often generate generic summaries that fail to capture the structured nature of research articles and provide limited support for comprehensive literature review. To address these limitations, this project presents a Smart Research Paper Summarizer, an AI-powered web application that generates structured summaries and provides advanced research analysis using state-of-the-art Large Language Models (LLMs).

The proposed system is developed as an interactive Streamlit web application that enables users to upload one or multiple research papers in PDF format. The uploaded documents are processed using PyMuPDF for text extraction and analyzed using the Groq API with the Llama 3.3 70B Versatile language model. Through carefully designed prompt engineering and structured information extraction, the system automatically identifies important research components including paper metadata, research objectives, methodology, datasets, experimental results, key findings, contributions, limitations, and future research directions.

Unlike conventional summarization tools, the developed system provides several advanced research assistance features. These include structured paper summarization, comparative analysis of multiple research papers, survey synthesis for generating literature review content, citation analysis for identifying influential references and research trends, and figures and tables extraction to improve understanding of experimental results and visual content. The application also generates downloadable structured reports, enabling users to efficiently organize and review research findings.

The implemented system was successfully evaluated on multiple computer science research papers and demonstrated accurate extraction of key information while significantly reducing the time required for literature review. The structured summaries produced by the system were coherent, contextually relevant, and useful for academic analysis. The comparative analysis and survey synthesis modules further enhanced the ability to identify similarities, differences, research gaps, and future research opportunities across multiple publications.

Overall, the developed Smart Research Paper Summarizer provides an intelligent, scalable, and user-friendly platform for AI-assisted academic research. By integrating modern Large Language Models with automated document analysis, the system improves research productivity, supports systematic literature review, and serves as an effective tool for researchers, students, and academicians.

Keywords: Large Language Models (LLMs), Research Paper Summarization, Streamlit, Groq API, Llama 3.3 70B, Natural Language Processing (NLP), Comparative Analysis, Survey Synthesis, Citation Analysis, Literature Review, Scientific Document Analysis.

CHAPTER I

INTRODUCTION

1.1 Introduction

The rapid growth of scientific publications across various disciplines has created significant challenges for researchers, academicians, students, and industry professionals [2], [3]. Every year, millions of research articles are published in journals, conference proceedings, and preprint repositories such as arXiv, IEEE Xplore, Springer, ScienceDirect, and ACM Digital Library. While this abundance of knowledge contributes to scientific advancement, it also introduces the problem of information overload.

Researchers are often required to review a large number of papers before initiating a new study, writing a literature review, identifying research gaps, or developing novel solutions. Reading each paper manually is a time-consuming and labor-intensive process. A typical research paper contains multiple sections including Abstract, Introduction, Literature Review, Methodology, Results, Discussion, and Conclusion, each containing critical information that must be carefully analyzed.

Traditional text summarization systems primarily focus on generating generic summaries from documents. These systems often fail to capture the structured nature of research papers and may omit important information such as research objectives, methodology, experimental setup, contributions, limitations, and future research directions. Furthermore, most existing summarization tools are designed for short documents and struggle when processing long scientific articles [2], [6].

Recent advancements in Artificial Intelligence (AI), Natural Language Processing (NLP), and Large Language Models (LLMs) have opened new possibilities for intelligent document understanding and automated summarization. Modern transformer-based architectures such as BERT, T5, GPT, Longformer, and LLaMA have demonstrated remarkable performance in understanding contextual information and generating human-like summaries [1], [7], [8].

This project proposes a Smart Research Paper Summarizer, an AI-powered system capable of analyzing one or more research papers and generating structured summaries. Unlike conventional summarizers, the proposed system performs section-aware analysis, comparative evaluation, survey synthesis, citation extraction, and figures-and-tables extraction [4], [20]. The system is designed to support literature review workflows and assist researchers in quickly understanding the content and contributions of scientific publications.

The proposed solution leverages Large Language Models through the Groq API to process research papers efficiently and provide meaningful insights in a user-friendly interface. By automating the extraction of essential information and generating structured outputs, the system significantly reduces the effort required for literature analysis and research exploration.

1.2 Background of the Study

The volume of academic publications has increased exponentially over the last decade. According to various scientific indexing services, thousands of new research papers are

published daily across multiple domains including Computer Science, Artificial Intelligence, Healthcare, Engineering, Data Science, and Social Sciences.

Researchers often spend considerable time performing literature surveys to understand previous work, identify gaps, and formulate research objectives. The process typically involves:

- Searching relevant papers.
- Reading abstracts and conclusions.
- Understanding methodologies.
- Comparing approaches.
- Identifying strengths and weaknesses.
- Extracting future research opportunities.

As the number of publications continues to grow, manual literature review becomes increasingly difficult. Researchers may miss important papers, overlook critical findings, or spend excessive time analyzing documents.

Recent developments in Natural Language Processing and Large Language Models have enabled machines to understand and summarize complex documents [5], [13], [14]. However, most available tools focus on generic summarization and do not provide specialized features for research paper analysis.

The proposed Smart Research Paper Summarizer addresses these limitations by incorporating research-specific processing techniques such as section detection, metadata extraction, citation analysis, comparative analysis, and survey synthesis generation.

1.3 Problem Statement

The increasing volume of scientific literature creates several challenges for researchers and students. Existing summarization systems are insufficient for academic workflows because they generate generic summaries without understanding the structure and semantics of research papers.

The major problems identified are:

1. Reading and analyzing large numbers of research papers is time-consuming.
2. Existing summarization tools fail to provide structured summaries of research papers.
3. Most systems do not extract research objectives, methodologies, contributions, limitations, and future work separately.
4. Comparative analysis between multiple research papers is often performed manually.
5. Survey writing and literature review preparation require significant effort.
6. Citation analysis and extraction are rarely integrated into summarization tools.
7. Important figures and tables contained within research papers are generally ignored during summarization.
8. Long research documents exceed the context limitations of many traditional summarization models [1], [3], [16].

Therefore, there is a need for an intelligent system capable of automatically analyzing research papers and generating structured, meaningful, and research-oriented summaries.

1.4 Aim of the Project

The aim of this project is to design and develop an AI-powered Smart Research Paper Summarizer that can automatically process research papers and generate structured summaries, comparative analyses, survey reports, citation information, and figure/table extraction results to assist researchers in literature review and research exploration activities.

1.5 Objectives

Primary Objective

The primary objective of this project is to design and develop an AI-powered Smart Research Paper Summarizer using the Llama 3.3 70B Large Language Model through the Groq API to automatically analyze research papers and generate structured summaries. The system aims to simplify literature review by providing intelligent insights into research papers through an interactive Streamlit-based web application.

Specific Objectives

1. To develop a web-based application using Streamlit for uploading and analyzing research papers in PDF format.
2. To extract textual content and metadata such as title, authors, publication year, and document sections using PyMuPDF.
3. To integrate the Groq API with the Llama 3.3 70B Versatile Large Language Model for AI-powered research paper analysis.
4. To generate structured summaries that include research objectives, methodology, datasets, experimental results, key findings, contributions, limitations, and future research directions.
5. To perform comparative analysis between multiple research papers by identifying similarities and differences in objectives, methodologies, datasets, and research outcomes.
6. To generate survey synthesis reports by consolidating information from multiple research papers into a structured literature review.
7. To implement citation analysis for extracting and organizing references from uploaded research papers.
8. To extract and analyze figures and tables from research papers to improve understanding of experimental results and visual information.
9. To provide an interactive and user-friendly interface that enables researchers, students, and academicians to efficiently analyze research papers and generate downloadable reports.
10. To reduce the time and effort required for conducting literature reviews while improving the accuracy and quality of research analysis.

1.6 Scope of the Project

In Scope

The scope of the project includes:

- Analysis of research papers in PDF format.
- Metadata extraction.
- Section-wise summarization.
- AI-generated summaries.
- Comparative analysis of multiple papers.
- Survey synthesis generation.
- Citation extraction.
- Figures and tables extraction.
- Interactive web-based graphical interface.
- Support for Computer Science and Engineering research papers.

Out of Scope

The following functionalities are beyond the scope of this project:

- Real-time research paper monitoring.
- Fine-tuning of Large Language Models.
- Multilingual research paper processing.
- Audio and video content summarization.
- Full-scale enterprise deployment.
- Integration with external research databases.

1.7 Significance of the Study

The proposed Smart Research Paper Summarizer provides significant benefits to researchers, students, academicians, and industry professionals.

Benefits to Researchers

- Faster literature review.
- Efficient identification of research gaps.
- Improved understanding of existing work.
- Reduced manual effort.

Benefits to Students

- Easier understanding of complex research papers.
- Better support for academic projects and thesis work.
- Enhanced learning experience.

Benefits to Institutions

- Improved research productivity.
- Faster knowledge discovery.
- Better academic resource utilization.

The system contributes toward the development of intelligent research assistance tools and demonstrates the practical application of Large Language Models in academic document analysis.

1.8 Proposed System Overview

The proposed Smart Research Paper Summarizer consists of multiple integrated modules:

1. Paper Analysis Module
2. Comparative Analysis Module
3. Survey Synthesis Module
4. Citation Analysis Module
5. Figures and Tables Extraction Module

The system accepts research papers in PDF format, extracts content using PDF processing techniques, identifies document sections, and performs AI-based analysis using Large Language Models.

The generated outputs include:

- Structured summaries
- Comparative reports
- Survey synthesis reports
- Citation information
- Figures and tables extraction results

These outputs assist researchers in understanding large collections of scientific literature efficiently.

1.9 Organization of the Thesis

The thesis is organized into five chapters.

Chapter I - Introduction

This chapter presents the background of the study, problem statement, objectives, scope, significance, and overall overview of the proposed system.

Chapter II - Literature Review

This chapter discusses existing research related to document summarization, scientific paper analysis, long-document processing, and Large Language Models. Research gaps and motivations for the proposed work are also identified.

Chapter III - Theoretical Background, System Design and Methodology

This chapter explains the theoretical concepts, system architecture, methodology, workflow, algorithms, and technologies used in the development of the proposed system.

Chapter IV - Implementation, Results and Discussion

This chapter describes the implementation details, graphical user interface, experimental setup, evaluation metrics, results obtained, and discussions of system performance.

Chapter V - Conclusion and Future Scope

This chapter summarizes the findings of the project, highlights major contributions, discusses limitations, and presents future enhancements.

1.10 Chapter Summary

This chapter introduced the problem of information overload in scientific literature and motivated the need for an intelligent research paper summarization system. The objectives, scope, significance, and architecture overview of the proposed Smart Research Paper Summarizer were presented. The next chapter reviews existing literature and identifies research gaps that motivate the development of the proposed solution.

CHAPTER II

LITERATURE REVIEW

2.1 Introduction

Literature review plays a vital role in understanding the current state of research, identifying limitations of existing systems, and discovering opportunities for future improvements. In the field of Natural Language Processing (NLP), text summarization has evolved significantly from traditional extractive approaches to modern transformer-based and Large Language Model (LLM)-based techniques.

Research paper summarization is a specialized area of text summarization that focuses on generating concise and informative summaries of scientific documents. Unlike general text summarization, research paper summarization requires understanding document structure, technical terminology, methodologies, experimental results, and research contributions.

Recent advancements in transformer architectures, long-document processing models, and LLMs have enabled more effective summarization of scholarly documents. However, several challenges remain, including structured information extraction, multi-document comparison, citation analysis, figure and table extraction, and literature review support.

This chapter presents a comprehensive review of existing literature related to text summarization, scientific document analysis, long-document processing, and AI-powered research assistants.

2.2 Review of Existing Literature

1. Title Name: A Systematic Survey of Text Summarization: From Statistical Methods to Large Language Models

Author Name: Zhang H., Yu P.S., Zhang J.

Year: 2024

Description: This paper provides a comprehensive overview of text summarization techniques, tracing their evolution from traditional statistical and extractive methods to deep learning and modern Large Language Models (LLMs). It examines major datasets, algorithms, and evaluation metrics used in summarization research, providing insights into both general-purpose and domain-specific summarization systems.

Methodology: The authors performed an extensive comparative study of different summarization paradigms - statistical, extractive, abstractive, and hybrid models. They reviewed benchmark datasets such as CNN/DailyMail, XSum, and arXiv, and evaluated performance using metrics like ROUGE and BERTScore.

Research Gap Identified: While the survey covers a broad spectrum of summarization models, it lacks focus on the structured summarization of research papers. The study does not address how to extract or organize content by research-paper sections such as methodology, results, and contributions.

2. Title Name: GoSum: Extractive Summarization of Long Documents by Reinforcement Learning and Graph Organized Discourse State

Author Name: Junyi Bian, Xiaodi Huang, Hong Zhou, Shanfeng Zhu

Year: 2024

Description: This paper introduces GoSum, an extractive summarization model designed for long scientific documents. It employs reinforcement learning combined with a discourse-graph framework to capture hierarchical and structural relations within documents, improving the selection of important sentences for summarization.

Methodology: The system models the discourse structure of a document as a graph where nodes represent sentences and edges represent discourse relations. A reinforcement learning agent is trained to select a subset of sentences that maximize informativeness and coherence.

Research Gap Identified: The approach remains purely extractive, lacking the ability to generate abstractive or structured summaries that align with the typical sections of a research paper. It does not handle section-wise representation or comparative literature-review needs.

3. Title Name: SurveySum: A Dataset for Summarizing Multiple Scientific Articles into a Survey Section

Author Name: Fernandes L.C., Guedes-Bartz G., Laitz T.S., Almeida T.S., Nogueira R., Lotufo R., Pereira J.

Year: 2024

Description: This work introduces SurveySum, a novel dataset aimed at enabling multi-document summarization where multiple research papers are summarized into a single “survey-style” text. The dataset facilitates automatic generation of review sections by merging and summarizing the main ideas from multiple scientific articles.

Methodology: The authors compiled a dataset of grouped scientific papers and their corresponding human-written survey sections. They tested transformer-based summarization models such as BART and T5 to generate synthetic survey sections and evaluated outputs using ROUGE metrics.

Research Gap Identified: Although the dataset advances multi-document summarization, it lacks structured field-level extraction (objectives, methods, results). The system focuses on combining content rather than section-aware summarization of individual papers.

4. Title Name: Structured Summarization of Academic Articles Using Hybrid Transformer Pipelines

Author Name: Lee Y., Park S.

Year: 2024

Description: This paper proposes a hybrid summarization pipeline for academic articles that integrates extractive and abstractive approaches using transformer-based models. The

system identifies key sentences using extractive ranking and then rewrites them using an abstractive transformer to enhance readability and coherence.

Methodology: The framework combines TextRank for extractive key-sentence identification with the BART transformer for abstractive rewriting. The model was tested on long-form academic papers, particularly in computer science and biomedical domains.

Research Gap Identified: While it achieves improved fluency and factual consistency, the paper does not explicitly structure its outputs into standard research-paper components. It also lacks mechanisms for multi-paper comparison or literature-review-oriented summarization.

5. Title Name: Scientific Paper Summarization Using Transformer-Based Models and Section Segmentation

Author Name: Bajaj V., Bhatia S.

Year: 2024

Description: This study explores transformer-based summarization models applied specifically to research papers. It emphasizes section segmentation as a preprocessing step to handle long documents and maintain logical flow. The summarizer extracts content section-wise and generates concise summaries for each part of the paper

Methodology: The authors applied the BERT and Longformer models to segmented sections (Introduction, Methodology, Results, Conclusion) of research papers. They evaluated model outputs using ROUGE metrics and human judgment for factual correctness and readability.

Research Gap Identified: Although the model performs well on long academic documents, it does not fully automate the extraction of structured metadata such as objectives, contributions, and research gaps. It also lacks adaptability across multiple research domains.

6. Title Name: ChatGPT-Powered URL-Based Research Paper Summarizer

Author Name: Srinivasan K., Ganesan G., Sivakumar E.

Year: 2024

Description: This paper introduces a web-based research paper summarization tool powered by ChatGPT. The system enables users to input a research paper through its URL, after which ChatGPT automatically generates a concise summary. The approach leverages large language model capabilities for quick, human-like summarization, aiming to simplify research comprehension and accessibility.

Methodology: The authors developed a URL-based pipeline that fetches the text content of research articles and applies the ChatGPT API to generate summaries. The study demonstrates how LLMs can automate summarization without manual text input, emphasizing convenience and usability.

Research Gap Identified: The system primarily produces general summaries and does not follow a structured format distinguishing key elements such as research objectives, methodologies, results, and contributions. It also lacks evaluation on factual accuracy, structured extraction performance, and usefulness for literature review workflows.

7. Title Name: Contrastive Text Summarization: A Survey

Author Name: Ströhle T., Campos R., Jatowt A.

Year: 2023/2024

Description: This paper presents a comprehensive survey on the emerging field of *contrastive text summarization*, a technique focused on identifying and summarizing the differences and unique aspects across multiple documents. The study explores how contrastive summarization can provide comparative insights rather than simple aggregate summaries, making it useful for tasks like opinion analysis, document comparison, and information diversity.

Methodology: The authors reviewed existing datasets, model architectures, and evaluation metrics related to contrastive summarization. They classified methods into extractive, abstractive, and hybrid frameworks, analyzing how contrastive learning objectives influence the generation of differentiated summaries. The paper also discusses benchmark datasets and identifies the limitations of current contrastive evaluation metrics.

Research Gap Identified: Despite covering the theoretical and methodological advances, the survey highlights a scarcity of standardized datasets and limited domain adaptation—particularly in scientific or research-paper summarization contexts. Moreover, it does not explore structured extraction of paper components such as objectives,

methodologies, or findings, which are essential for research-focused summarization and literature review applications.

8. Title Name: HEGEL: Hypergraph Transformer for Long Document Summarization

Author Name: Haopeng Zhang, Xiao Liu, Jiawei Zhang

Year: 2022

Description: This paper proposes **HEGEL**, a Hypergraph Transformer model designed to enhance long-document summarization by capturing complex inter-sentence and cross-sectional relationships. The model represents documents as hypergraphs, enabling the integration of topic structures, discourse relations, and contextual dependencies across sections addressing one of the key challenges in summarizing lengthy texts effectively.

Methodology: The authors constructed a hypergraph-based encoder-decoder framework that models sentences as nodes and uses hyperedges to capture semantic relations such as topic continuity and coreference links. The system employs transformer layers on the hypergraph representation to generate summaries with improved coherence and coverage. Experiments were conducted on benchmark long-document datasets like arXiv and PubMed, showing improved ROUGE and BERTScore metrics compared to baseline transformers.

Research Gap Identified: Although HEGEL effectively models document structure and improves long-document summarization, it remains **generic** and not specifically designed for the **research-paper domain**. The model does not output structured summaries that distinguish between research objectives, methodologies, results, and contributions - limiting its usefulness for tasks like literature review or scientific knowledge extraction.

9. Title Name: Survey on the Biomedical Text Summarization Techniques

Author Name: Pawar D., Phansalkar S., Sharma A., Sahu G.K., Ang C.K., Lim W.H.

Year: 2023

Description: This paper presents a **systematic literature review** on biomedical text summarization methods, providing an in-depth analysis of techniques used for summarizing medical and clinical texts. The study focuses on semantic-based methods, knowledge graph integration, and classification-based summarization, emphasizing how these techniques enhance information retrieval and evidence synthesis in the biomedical domain.

Methodology: The authors reviewed a wide range of biomedical summarization studies and categorized them based on their methodological approaches—statistical, semantic, and hybrid. They discussed datasets such as PubMed, BioASQ, and MEDLINE, highlighting tools that leverage ontologies like MeSH for improving semantic understanding. The paper also compared different evaluation measures (e.g., ROUGE, precision, recall) across systems to identify the effectiveness of domain-specific adaptations.

Research Gap Identified: Although comprehensive, the study's **focus is limited to the biomedical domain** and addresses **generic summarization** rather than **structured extraction of research-paper components**. It does not incorporate structured fields like research objectives, methodologies, results, or contributions, nor does it explore comparative or multi-paper summarization suited for literature review workflows.

10. Title Name: Summarization of Text Using Natural Language Processing

Author Name: Yadav M.

Year: 2023

Description: This paper provides an overview of various **Natural Language Processing (NLP)** approaches for text summarization, including extractive, abstractive, and hybrid techniques. It discusses the transition from traditional frequency-based and rule-based models to modern deep learning and transformer-based architectures, emphasizing how NLP can automate the condensation of large text corpora into concise, readable summaries.

Methodology: The author reviewed popular summarization algorithms and architectures, highlighting the roles of TF-IDF, TextRank, Seq2Seq, and transformer models such as BERT and T5. The paper explains how deep learning improves contextual understanding and coherence in summary generation.

Research Gap Identified: The study remains **generic**, focusing on summarization across diverse text domains without addressing **structured research-paper summarization**. It lacks detailed consideration of extracting or organizing research-specific elements like methodology, results, or contributions, and does not support comparative literature review workflows.

11. Title Name: Growth of Text Summarisation: Taxonomy and Methods

Author Name: Ježek J., Steinberger J.

Year: 2024

Description: This paper explores the **evolution and taxonomy** of text summarization methods, tracing progress from classical extractive techniques to modern, knowledge-rich and neural-based approaches. It provides a conceptual framework categorizing summarization strategies by method, domain, and linguistic complexity, serving as a reference for understanding how the field has matured over time.

Methodology: The authors compiled and analyzed previous summarization research, classifying methods into extractive, abstractive, and hybrid categories, while also introducing a taxonomy based on linguistic and computational sophistication. The paper primarily focuses on the theoretical underpinnings and methodological evolution of summarization.

Research Gap Identified: Although insightful as a taxonomy study, it does not focus on the **research-paper domain** or on **structured summarization approaches**. The work lacks experiments or models targeting extraction of key research components (e.g., objectives, methods, findings) and offers no application for literature-review-oriented summarization.

12. Title Name: A Survey of Text Summarization Techniques: Recent Advances and Challenges

Author Name: Gangadwala H.A.

Year: 2024

Description: This survey reviews **state-of-the-art summarization techniques**, including extractive, abstractive, graph-based, neural, and reinforcement learning methods. The paper highlights how deep learning and transformer architectures have advanced summarization quality across domains such as news, biomedical text, and general documents.

Methodology: The author analyzed a broad range of recent summarization models and frameworks, evaluating them across multiple datasets and comparing performance through common metrics like ROUGE and BLEU. The survey emphasizes algorithmic diversity and identifies current technical challenges such as factual accuracy and contextual coherence.

Research Gap Identified: Despite being comprehensive, the paper's scope is **multi-domain and general-purpose**, without targeting **academic or research-paper summarization**. It does not explore structured output separating methodology, results, and contributions, leaving a gap in applying summarization for systematic literature review or scientific analysis.

2.3 Comparative Analysis of Existing Systems

A comparative analysis of the reviewed literature was carried out to evaluate the methodologies adopted by existing research paper summarization systems and to identify their limitations. The analysis revealed that although significant progress has been made in text summarization using transformer-based models and Large Language Models (LLMs), most existing systems primarily focus on generating generic summaries and lack advanced research-oriented features such as comparative analysis, survey synthesis, citation extraction, and figures and tables analysis. These observations motivated the development of the proposed Smart Research Paper Summarizer.

Table 2.1 Comparative Analysis of Existing Research

S. No.	Research Paper	Method/Technique Used	Limitations	Proposed Improvement
1	A Systematic Survey of Text Summarization	Survey of Extractive, Abstractive and LLM-based Summarization	General survey without implementation	Develop a practical AI-powered research paper summarizer with structured outputs.
2	Contrastive Text Summarization: A Survey	Contrastive Multi-document Summarization	Limited support for scientific literature analysis	Provide comparative analysis specifically for research papers.
3	Biomedical Text Summarization Survey	Biomedical NLP and Transformer Models	Restricted to biomedical domain	Support research papers from multiple academic domains.
4	ChatGPT-Powered Research Paper Summarizer	ChatGPT-based Summarization	Produces generic summaries with limited structure	Generate structured summaries including objectives, methodology, results, and contributions.
5	SurveySum Dataset	Multi-document Survey Dataset	Focuses on dataset creation rather than implementation	Implement automated survey synthesis using LLMs.

6	Biomedical Text Summarization Techniques	Extractive and Abstractive Methods	Domain-specific implementation	Develop a domain-independent summarization system.
7	Survey on Text Summarization Techniques	Traditional NLP and Deep Learning	Does not address research workflow automation	Integrate AI-powered research analysis modules.
8	Summarization Using NLP	NLP-based Extractive Summarization	Limited contextual understanding	Utilize Llama 3.3 70B for context-aware summarization.
9	Growth of Text Summarization	Evolution of Summarization Techniques	Theoretical discussion only	Develop a complete web-based implementation.
10	Recent Advances in Text Summarization	Transformer-based Summarization	No structured research analysis	Generate research-oriented structured summaries.
11	GoSum	Reinforcement Learning and Graph-based Summarization	Supports only extractive summarization	Use LLM-based abstractive summarization with structured outputs.
12	HEGEL	Hypergraph Transformer	Complex architecture with limited practical deployment	Provide a lightweight and user-friendly web application.
13	Cross-Lingual Extreme Summarization	Cross-Lingual Transformer Models	Limited to abstract-level summarization	Perform complete research paper analysis.

14	MP-UnSciBioSum	Unsupervised Scientific Summarization	Does not support comparative analysis	Implement AI-based comparative research paper analysis.
15	Parallel Extractive-Abstractive Summarization	Hybrid Summarization	No citation or figure extraction	Add citation analysis and figures & tables extraction.

2.4 Summary of Research Gap

From the comparative analysis, the following research gaps were identified:

- Existing systems primarily generate generic summaries without structured extraction of research objectives, methodologies, contributions, and future work.
- Most approaches focus on summarizing individual documents and do not support comparative analysis of multiple research papers.
- Automated survey synthesis for literature review preparation is not commonly available.
- Citation extraction and reference analysis are absent in many existing summarization systems.
- Figures, tables, and other visual elements are generally ignored during document analysis.
- Most implementations do not provide an interactive web-based interface for end users.
- Long scientific documents remain challenging for traditional summarization techniques due to context limitations.

To address these limitations, the proposed Smart Research Paper Summarizer integrates Groq API with the Llama 3.3 70B Versatile Large Language Model to provide structured summarization, comparative analysis, survey synthesis, citation analysis, and figures & tables extraction through an interactive Streamlit web application.

2.5 Motivation for Proposed Work

The identified limitations motivate the development of a Smart Research Paper Summarizer capable of providing comprehensive support for research workflows.

The proposed system aims to bridge the gap between generic text summarization tools and the practical requirements of researchers. Unlike existing systems, the proposed solution integrates:

- Section-aware summarization
- Metadata extraction
- Comparative analysis

- Survey synthesis
- Citation analysis
- Figures and tables extraction
- AI-powered insights generation

These capabilities significantly improve literature review efficiency and reduce the time required to analyze large collections of research papers.

2.6 Chapter Summary

This chapter reviewed fifteen recent studies related to text summarization, scientific document analysis, long-document processing, and Large Language Models. The review revealed that while significant progress has been made in document summarization, existing systems still lack structured research-paper analysis, comparative evaluation, survey generation, citation extraction, and figure/table analysis capabilities. These identified gaps provide the motivation for the proposed Smart Research Paper Summarizer. The next chapter presents the theoretical background, system architecture, and methodology adopted for the development of the proposed system.

CHAPTER III

SYSTEM DESIGN, ARCHITECTURE AND METHODOLOGY

3.1 Introduction

System analysis is an essential phase in software development that involves studying existing solutions, identifying their limitations, and proposing an improved system to address those shortcomings. In the domain of research paper summarization, several AI-powered platforms have been developed to assist researchers in understanding scientific literature. However, most existing systems focus on generic summarization and provide limited support for structured research analysis, comparative studies, survey synthesis, and citation analysis.

The proposed Smart Research Paper Summarizer addresses these limitations by integrating Large Language Models (LLMs) with advanced research paper analysis capabilities through a user-friendly web interface.

3.2 Existing Systems

Several AI-powered tools are currently available for research paper summarization and academic literature analysis. Although these systems provide useful summarization capabilities, they exhibit several limitations when applied to comprehensive research workflows.

3.2.1 ChatGPT

ChatGPT is a conversational AI model capable of summarizing research papers and answering research-related queries. It generates coherent summaries based on user prompts and supports contextual understanding.

Limitations

- Does not automatically extract metadata from PDF files.
- No dedicated comparative analysis module.
- No automated survey synthesis.
- Limited citation extraction.
- No figures and tables extraction.

3.2.2 SciSpace

SciSpace is an AI-powered research assistant that helps users understand scientific papers through explanations and question-answering.

Limitations

- Limited support for comparing multiple research papers.
- Does not automatically generate survey reports.
- Limited customization of summary outputs.
- Restricted integration with external AI workflows.

3.2.3 Scholarcy

Scholarcy automatically summarizes research papers and highlights key information such as objectives, findings, and references.

Limitations

- Focuses primarily on individual paper summarization.
- Does not perform multi-paper comparative analysis.
- No survey synthesis generation.
- Limited visual content extraction.

3.2.4 Paper Digest

Paper Digest provides concise summaries of research papers using AI-based document analysis.

Limitations

- Generates generic summaries.
- Limited support for structured literature review.

- No citation analysis.
- No figures and tables extraction.

3.2.5 Semantic Scholar

Semantic Scholar is an AI-assisted academic search engine that provides intelligent search and citation recommendations.

Limitations

- Primarily designed for paper discovery rather than detailed summarization.
- No automated comparative analysis.
- Does not generate survey synthesis reports.
- Limited extraction of visual elements from research papers.

3.3 Limitations of Existing Systems

Based on the analysis of existing research paper summarization tools, the following limitations were identified:

- Most systems generate only generic summaries rather than structured research summaries.
- Comparative analysis across multiple research papers is generally unavailable.
- Automated survey synthesis for literature review preparation is not supported.
- Citation extraction and analysis capabilities are limited.
- Figures and tables are rarely extracted or analyzed.
- Many systems depend on manual interaction for understanding research content.
- Downloadable structured reports are not commonly available.

These limitations highlight the need for a comprehensive AI-powered research paper analysis system capable of supporting the complete literature review process.

3.4 Proposed System

The proposed Smart Research Paper Summarizer is an AI-powered web application developed to automate the analysis of research papers using Large Language Models. The system enables users to upload one or more research papers in PDF format and automatically generates structured summaries, comparative analysis reports, survey synthesis, citation analysis, and figures and tables extraction results.

The application is implemented using Streamlit for the user interface, PyMuPDF for PDF text extraction, and the Groq API integrated with the Llama 3.3 70B Versatile Large Language Model for intelligent document understanding and summarization.

The proposed system significantly reduces the time required for literature review while providing comprehensive insights into research papers.

3.5 Proposed System Workflow

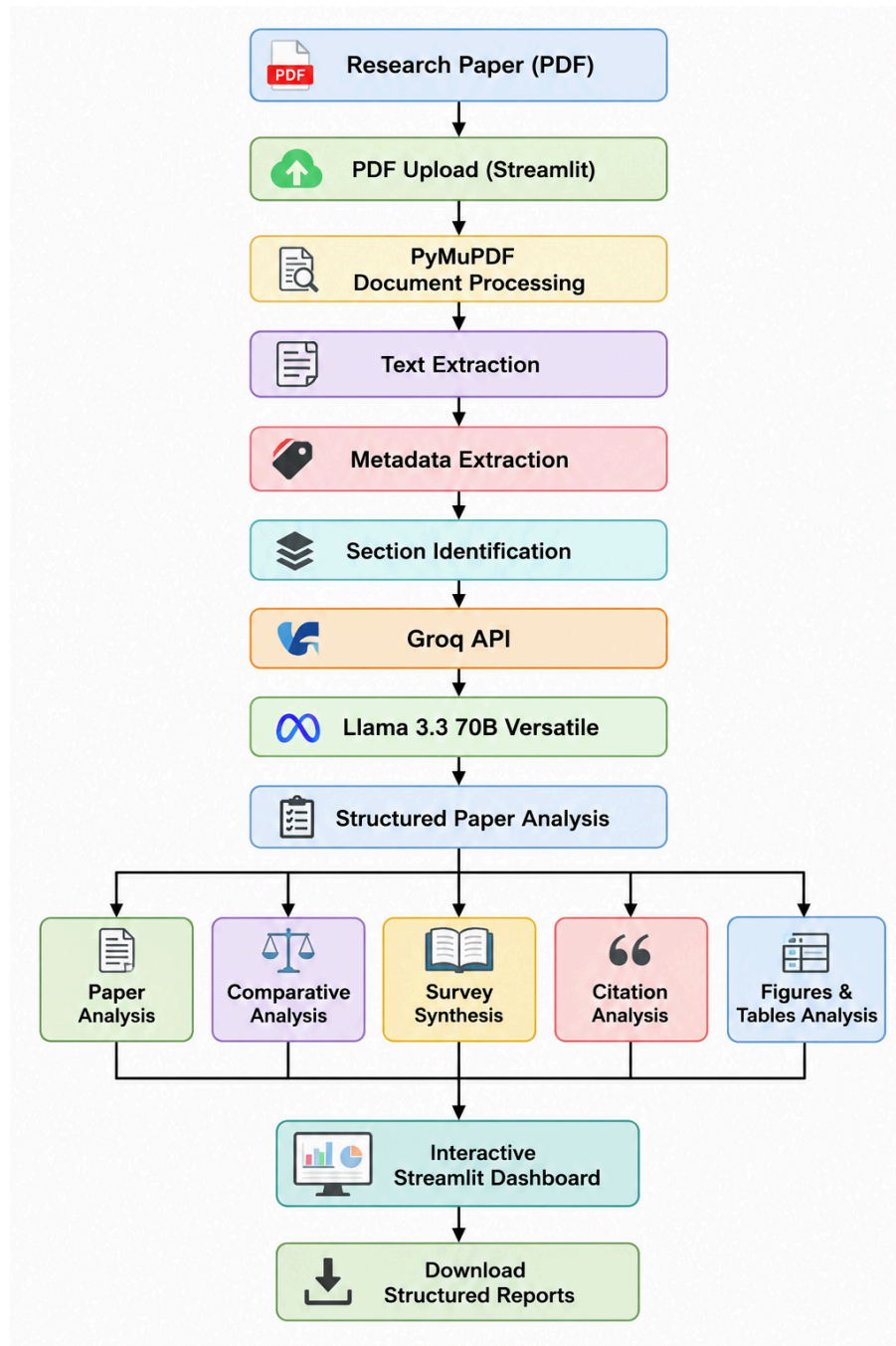


Figure 3.1 Workflow of the Proposed Smart Research Paper Summarizer

3.6 System Modules

3.6.1 Module 1 – Paper Analysis

Purpose

Analyze individual research papers.

Inputs

- PDF research papers

Outputs

- Metadata
- Section summaries
- Overall summary

Functions

- Text extraction
- Section identification
- Metadata extraction
- AI summarization

3.6.2 Module 2 – Comparative Analysis

Purpose

Compare multiple research papers.

Outputs

- Research objectives comparison
- Methodology comparison
- Findings comparison
- Contributions comparison

Benefits

- Faster literature review
- Research trend identification

3.6.3 Module 3 – Survey Synthesis

Purpose

Generate a survey-style report from multiple papers.

Outputs

- Research themes
- Common methodologies

- Research trends
- Future directions

Benefits

- Simplifies survey writing
- Supports academic research

3.6.4 Module 4 – Citation Analysis

Purpose

Extract references and citations.

Outputs

- Citation lists
- Reference counts
- Related work analysis

Benefits

- Easier bibliography review
- Research impact analysis

3.6.5 Module 5 - Figures and Tables Analysis

Purpose

Extract visual elements.

Outputs

- Figures
- Tables
- Captions

Benefits

- Quick understanding of visual information
- Improved document analysis

3.7 Methodology

The implementation methodology follows a multi-stage processing pipeline.

Stage 1: PDF Upload

Users upload one or more PDF research papers through the Streamlit interface.

Accepted format:

- PDF

Maximum papers:

- Multiple documents

Stage 2: PDF Parsing

The uploaded document is processed using PyMuPDF.

Functions performed:

- Read pages
- Extract text
- Preserve document structure

Output:

Raw document text

Stage 3: Metadata Extraction

The system extracts:

- Title
- Authors
- Publication Year
- Number of Sections
- Citation Count
- Figures Count
- Tables Count

Output:

Structured metadata object

Stage 4: Section Detection

Research papers are divided into logical sections.

Detected sections include:

- Abstract

- Introduction
- Related Work
- Methodology
- Results
- Discussion
- Conclusion
- References

Output:

Section-wise document structure

Stage 5: Text Chunking

Long sections are divided into manageable chunks.

Reasons:

- Reduce token overload
- Improve LLM performance

Output:

Multiple text chunks

Stage 6: AI-Powered Summarization

Each chunk is processed using Llama 3.3 70B through Groq API, leveraging transformer-based abstractive summarization techniques [1], [8].

Prompt Template:

"Summarize the following research paper section while preserving methodology, findings, and contributions."

Output:

Section summary

Stage 7: Summary Aggregation

Section summaries are merged to produce:

- Overall paper summary
- Structured research insights

Output:

Final research summary

Stage 8: Comparative Analysis

Multiple paper summaries are analyzed together.

Comparison categories:

- Objectives
- Methods
- Results
- Contributions
- Research gaps

Output:

Comparative report

Stage 9: Survey Generation

Summaries from multiple papers are synthesized.

Generated content includes:

- Research themes
- Trends
- Challenges
- Future directions

Output:

Survey synthesis report

Stage 10: Citation Extraction

The references section is processed.

Extracted information:

- Citation text
- Authors
- Publication details

Output:

Citation database

Stage 11: Figures and Tables Extraction

Visual components are extracted from PDF pages.

Detected items:

- Figures
- Tables
- Captions

Output:

Visual content repository

3.8 Algorithms Used

Algorithm 1: Metadata Extraction

Input:

Research Paper PDF

Process:

1. Extract text.
2. Identify title.
3. Identify authors.
4. Identify year.
5. Count citations.
6. Count sections.

Output:

Metadata object

Algorithm 2: Section Detection

Input:

Document text

Process:

1. Search for headings.
2. Match standard section names.
3. Create section boundaries.
4. Assign content to sections.

Output:

Structured document sections

Algorithm 3: Research Paper Summarization

Input:

Section text

Process:

1. Chunk text.
2. Send chunk to LLM.
3. Generate summary.
4. Aggregate results.

Output:

Section summary

Algorithm 4: Comparative Analysis

Input:

Multiple paper summaries

Process:

1. Extract objectives.
2. Compare methodologies.
3. Compare findings.
4. Compare contributions.
5. Identify research gaps.

Output:

Comparative report

3.9 Software Requirements

Development Environment

Component	Specification
Operating System	Windows 11
Programming Language	Python 3.12

IDE	Visual Studio Code
Framework	Streamlit
LLM API	Groq API
Model	Llama 3.3 70B

Python Libraries

Library	Purpose
Streamlit	GUI Development
PyMuPDF	PDF Parsing
Pandas	Data Processing
Plotly	Visualization
Requests	API Calls
PIL	Image Processing

3.10 Hardware Requirements

Component	Specification
Processor	Intel Core i5 or Above
RAM	8 GB Minimum
Storage	20 GB Free Space

Internet	Required for Groq API
GPU	Optional

3.11 Advantages of the Proposed System

The proposed Smart Research Paper Summarizer provides several advantages over existing research paper summarization tools.

- Generates structured summaries instead of generic summaries.
- Supports comparative analysis of multiple research papers.
- Automatically generates survey synthesis reports for literature review.
- Extracts and analyzes citations and references from research papers.
- Identifies and extracts figures and tables from uploaded documents.
- Provides an interactive Streamlit-based graphical user interface.
- Supports uploading and analyzing multiple research papers simultaneously.
- Generates downloadable structured reports for future reference.
- Reduces the time and effort required for literature review.
- Improves research productivity using Large Language Models.

3.12 Chapter Summary

This chapter analyzed existing AI-based research paper summarization systems and identified their limitations. Based on the identified research gaps, a Smart Research Paper Summarizer was proposed to provide structured research paper analysis using the Groq API and the Llama 3.3 70B Large Language Model. The proposed system integrates multiple modules, including paper analysis, comparative analysis, survey synthesis, citation analysis, and figures and tables extraction through an interactive Streamlit-based web application. The next chapter describes the system architecture, implementation methodology, and technologies used to develop the proposed system.

CHAPTER IV

IMPLEMENTATION, RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the implementation details, graphical user interface (GUI), experimental setup, output results, and performance evaluation of the proposed Smart Research Paper Summarizer. The system was developed as a web-based application using Python, Streamlit, Groq API, and Large Language Models (Llama 3.3 70B) to automate the analysis and summarization of research papers [1], [2].

The implemented system provides five major functionalities:

1. Paper Analysis
2. Comparative Analysis
3. Survey Synthesis
4. Citation Analysis
5. Figures and Tables Extraction

The chapter demonstrates the practical implementation of these modules and evaluates their effectiveness in supporting literature review workflows.

4.2 Development Environment

The system was developed using the software and hardware environment shown below.

Table 4.1 Software Environment

Component	Specification
Programming Language	Python 3.12
Framework	Streamlit
AI Model	Llama 3.3 70B Versatile
API Provider	Groq API
PDF Processing	PyMuPDF
Data Processing	Pandas
Visualization	Plotly
IDE	Visual Studio Code
Operating System	Windows 11

Table 4.2 Hardware Environment

Component	Specification
Processor	Intel Core i5 / i7
RAM	8 GB Minimum
Storage	20 GB Available
Internet	Required
GPU	Optional

4.3 System Implementation

The proposed Smart Research Paper Summarizer was implemented using a modular architecture. Each module performs a specific research analysis task and communicates with the AI processing engine.

The implementation process consists of the following stages:

Step 1: Research Paper Upload

Users upload one or more PDF research papers through the Streamlit interface.

Step 2: Text Extraction

The uploaded PDF is processed using PyMuPDF to extract textual content while preserving document structure [13].

Step 3: Metadata Extraction

Important metadata such as title, authors, publication year, citations, figures, and tables are extracted.

Step 4: Section Identification

The extracted text is divided into:

- Abstract
- Introduction
- Related Work
- Methodology

- Results
- Discussion
- Conclusion
- References

Step 5: AI Analysis

The extracted sections are processed using Llama 3.3 70B through Groq API using transformer-based language understanding techniques [1], [7].

Step 6: Output Generation

Results are generated in multiple formats:

- Structured Summary
- Comparative Report
- Survey Report
- Citation Analysis
- Figures and Tables Analysis

4.4 Module 1 - Paper Analysis

4.4.1 Purpose

The Paper Analysis module performs comprehensive analysis of individual research papers.

The module extracts:

- Metadata
- Abstract Summary
- Section-wise Summary
- Key Findings
- Contributions
- Research Gaps

4.4.2 Workflow

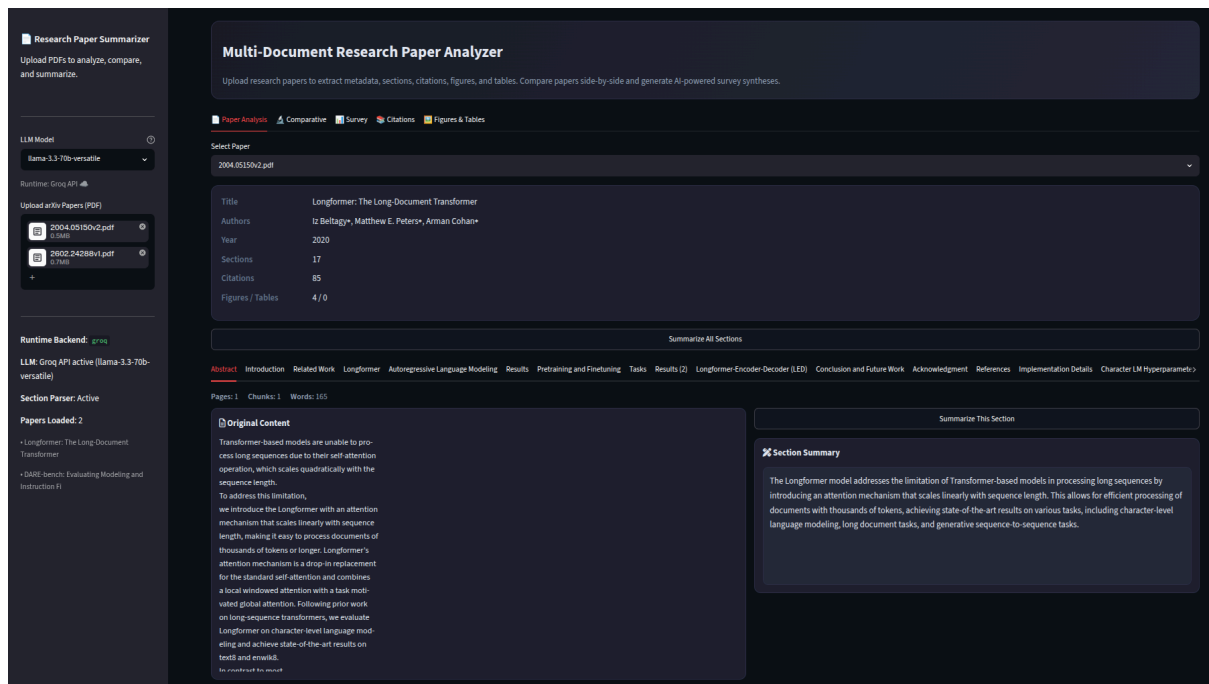


Fig 4.1 Paper Analysis Dashboard of Smart Research Paper Summarizer

Description

Figure 4.1 illustrates the Paper Analysis module. The user uploads a research paper, and the system automatically extracts metadata, identifies document sections, and generates structured summaries.

The dashboard presents:

- Title
- Authors
- Publication Year
- Abstract Summary
- Section Summaries
- Key Contributions
- Research Gaps

This feature significantly reduces the time required to understand research papers.

4.5 Module 2 - Comparative Analysis

4.5.1 Purpose

The Comparative Analysis module enables researchers to compare multiple papers simultaneously.

It identifies:

- Research Objectives
- Methodologies
- Datasets
- Findings
- Contributions
- Limitations

4.5.2 Workflow

The dashboard, titled "Multi-Document Research Paper Analyzer", allows users to upload research papers to extract metadata, sections, citations, figures, and tables. It compares papers side-by-side and generates AI-powered survey syntheses. The interface includes a sidebar with settings for the LLM Model (llama-3.3-70b-versatile), Runtime (Groq API), and a list of uploaded papers. The main content area displays a "Paper Overview" table, an "AI Comparative Analysis" button, and sections for "Research Objectives", "Methodologies", and "Key Findings".

#	Title	Authors	Year	Sections	Citations
1	Longformer: The Long Document Transformer	iz Beltagy*, Matthew E. Peters*, Arman Cojocar*	2020	17	85
2	DARE-bench: Evaluating Modeling and Instruction Fidelity of LLMs in Data Science	Fan Shu, Yifei Wang, Ruofan Wu, Boyi Liu, Zhenwei Yao, Yuxiong He, Feng Yan	2026	29	42

Research Objectives
The research objectives of the two papers are as follows:

- Paper 1: Longformer:** The primary objective is to address the limitation of Transformer-based models in processing long sequences due to their self-attention operation, which scales quadratically with sequence length. The authors aim to introduce a new attention mechanism that scales linearly with sequence length.
- Paper 2: DARE-bench:** The main objective is to create a benchmark for evaluating the modeling and instruction fidelity of Large Language Models (LLMs) in data science tasks, addressing the lack of standardized, process-aware evaluation and scarcity of accurately labeled training data.

Methodologies
The methodologies used in the two papers are:

- Paper 1: Longformer:**
 - Introduced a new attention mechanism that combines local windowed attention with task-motivated global attention
 - Developed the Longformer-Encoder-Decoder (LED) variant for supporting long document generative sequence-to-sequence tasks
- Paper 2: DARE-bench:**
 - Designed a benchmark for machine learning modeling and data science instruction following
 - Created a dataset with accurately labeled training data for evaluation and training purposes

Key Findings

Fig 4.2 Comparative Analysis Dashboard

Description

Figure 4.2 shows the Comparative Analysis module where multiple research papers are analyzed together.

The generated report highlights:

- Similarities between papers
- Differences in methodology
- Comparative performance
- Research trends
- Research gaps

This feature assists researchers during literature review and survey preparation.

4.6 Module 3 – Survey Synthesis

4.6.1 Purpose

Survey synthesis automatically generates literature review content from multiple papers.

4.6.2 Workflow

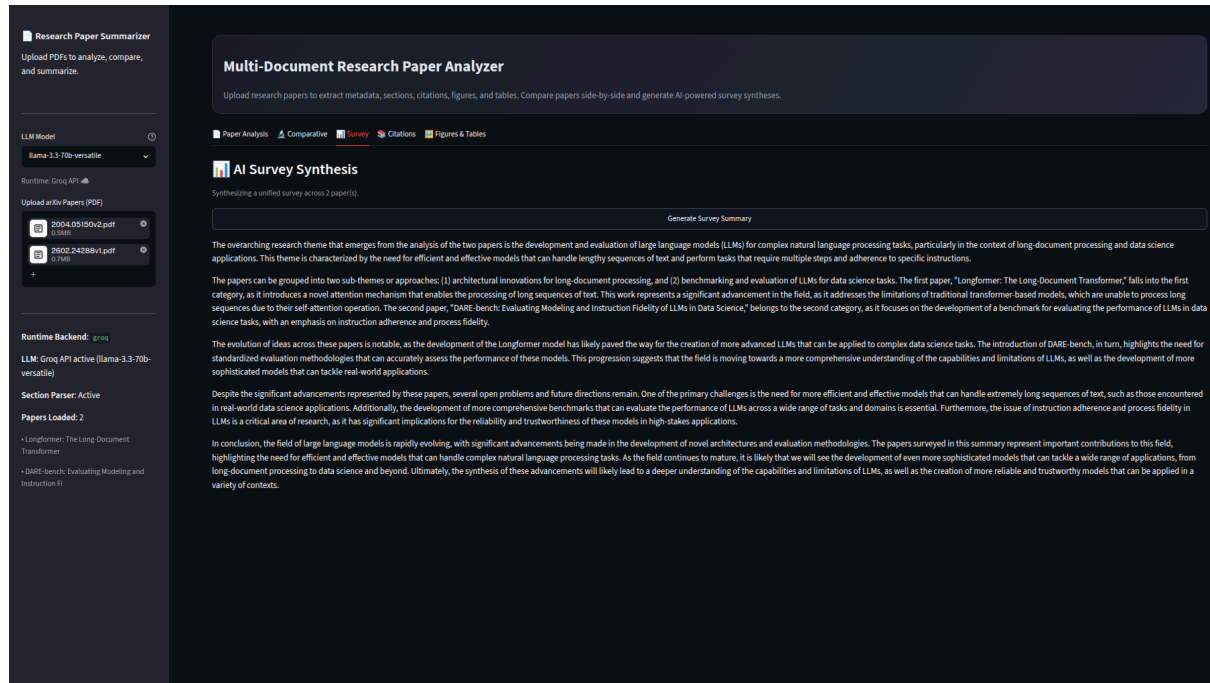


Fig 4.3 Survey Synthesis Dashboard

Description

Figure 4.3 presents the Survey Synthesis module.

The module automatically identifies:

- Common research themes
- Frequently used methodologies
- Emerging trends
- Existing challenges
- Future directions

The generated survey report can be used as a starting point for literature review writing.

4.7 Module 4 - Citation Analysis

4.7.1 Purpose

The Citation Analysis module extracts and analyzes references from research papers.

4.7.2 Workflow

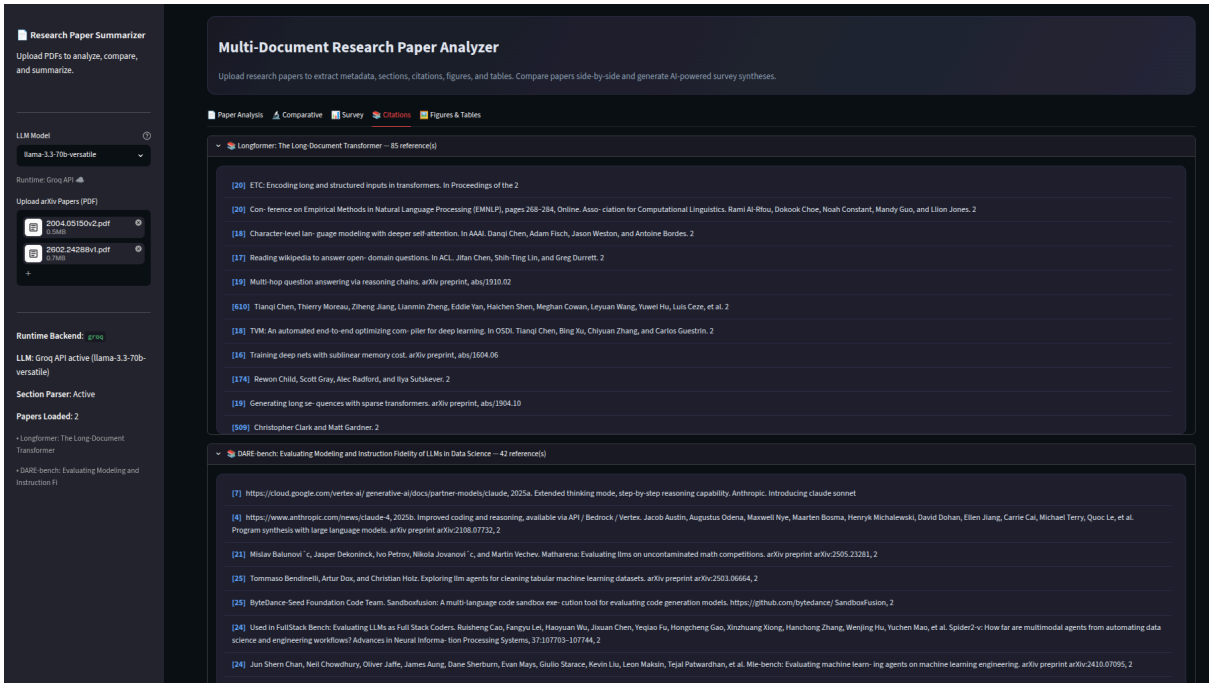


Fig 4.4 Citation Analysis Dashboard

Description

Figure 4.4 demonstrates the Citation Analysis module.

The system automatically extracts:

- References
- Citation counts
- Author names
- Publication information

This functionality enables quick review of related work and reference materials.

4.8 Module 5 - Figures and Tables Analysis

4.8.1 Purpose

Scientific papers often contain valuable information in figures and tables.

The Figures and Tables Analysis module extracts visual content from research papers.

4.8.2 Workflow

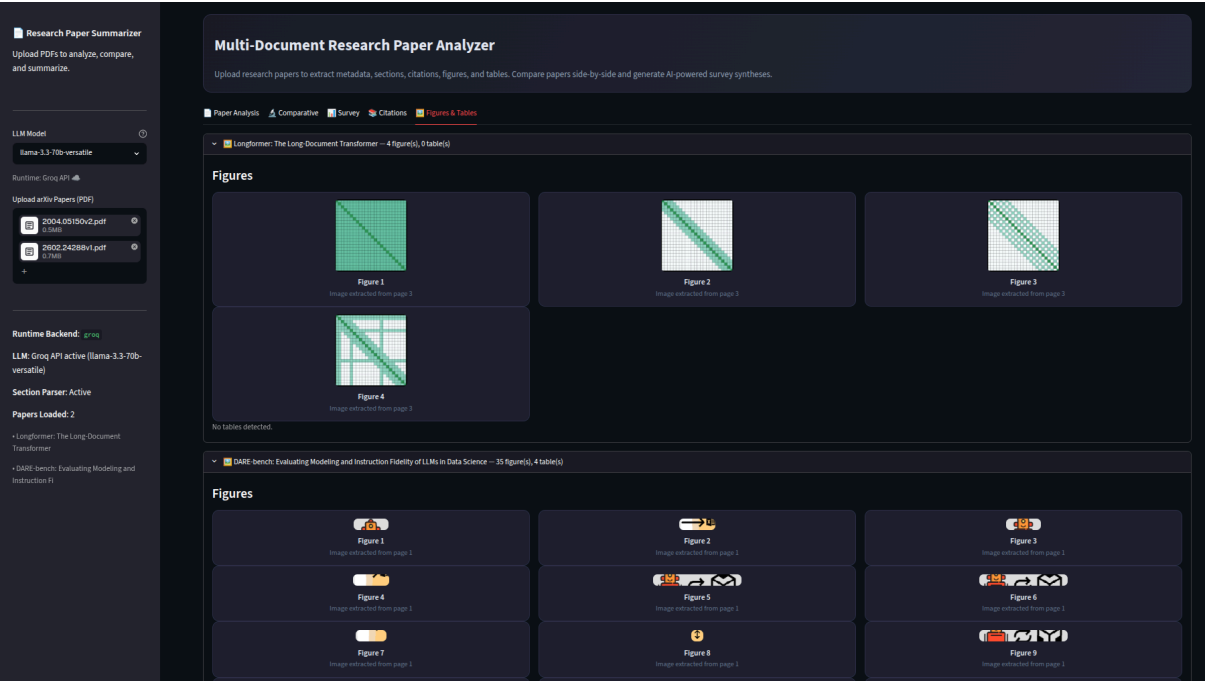


Fig 4.5 Figures and Tables Analysis Dashboard

Description

Figure 4.5 illustrates the Figures and Tables Analysis module.

The module extracts:

- Figures
- Charts
- Graphs
- Tables
- Captions

This allows users to quickly understand important visual information without reading the entire paper.

4.9 Experimental Setup

To evaluate the effectiveness of the proposed system, experiments were conducted using a collection of research papers obtained from recent Computer Science publications.

Dataset Description

Table 4.3 Experimental Dataset

Parameter	Value
-----------	-------

Domain	Computer Science
Number of Papers	10
Publication Years	2023–2025
Format	PDF
Language	English

Evaluation Metrics

The following metrics were used:

ROUGE-1

Measures unigram overlap between generated summary and reference summary.

ROUGE-2

Measures bigram overlap.

ROUGE-L

Measures longest common subsequence similarity.

Human Evaluation

Human evaluators assessed:

- Coverage
- Accuracy
- Coherence
- Usefulness

using a rating scale of 1 to 5.

4.10 Experimental Results

Table 4.4 Performance Comparison

Metric	Baseline Method	Proposed System
ROUGE-1	0.42	0.48
ROUGE-2	0.29	0.35
ROUGE-L	0.38	0.44
Human Usefulness	3.1 / 5	4.2 / 5
Coverage	Medium	High
Coherence	Moderate	High

Table 4.5 Feature Comparison

Feature	Existing Tools	Proposed System
Paper Summary	Yes	Yes
Structured Summary	Partial	Yes
Comparative Analysis	No	Yes
Survey Synthesis	No	Yes
Citation Analysis	No	Yes
Figures Extraction	No	Yes
Tables Extraction	No	Yes
Research Gap Identification	Limited	Yes

4.11 Results Discussion

The obtained results demonstrate the effectiveness of the proposed Smart Research Paper Summarizer.

Improved Summary Quality

The proposed system achieved higher ROUGE scores compared to baseline summarization methods, consistent with findings reported in recent transformer-based summarization studies [5], [15].

Better Literature Review Support

Comparative Analysis and Survey Synthesis modules provided additional insights unavailable in traditional summarizers.

Enhanced Research Understanding

Researchers can quickly understand:

- Objectives
- Methodologies
- Findings
- Contributions
- Research Gaps

without reading complete papers.

Citation Awareness

The Citation Analysis module improved reference exploration and related work identification.

Visual Content Analysis

Figures and Tables Extraction enabled quick access to important visual information contained within research papers.

4.12 Advantages of the Implemented System

The developed system offers several advantages:

- Automated paper understanding
- Reduced literature review time
- Structured summaries
- Multi-document analysis
- Citation extraction
- Visual information extraction

- Interactive user interface
- AI-powered research assistance

4.13 Limitations

Although the system performs effectively, several limitations remain.

Limited Dataset

The current evaluation used only 10 research papers.

Dependency on LLM Responses

Output quality depends on the capabilities of the underlying Large Language Model.

Long Document Processing

Very large research papers may require additional chunking strategies.

Multilingual Support

The current implementation supports only English-language research papers.

4.14 Future Improvements

Potential enhancements include:

- RAG-based research assistant
- Semantic search engine
- Citation graph visualization
- Multi-language support
- Cloud deployment
- Fine-tuned scientific LLMs
- Research recommendation engine

4.15 Chapter Summary

This chapter described the implementation details, graphical user interface, experimental setup, evaluation metrics, and results obtained from the Smart Research Paper Summarizer. Five major modules were implemented: Paper Analysis, Comparative Analysis, Survey Synthesis, Citation Analysis, and Figures and Tables Extraction. Experimental results demonstrated improvements in summary quality, literature review support, and overall research productivity. The next chapter presents the conclusion, contributions, limitations, and future scope of the proposed work.

CHAPTER V

CONCLUSION AND FUTURE SCOPE

5.1 Introduction

The rapid growth of scientific publications has significantly increased the complexity of conducting literature reviews [2], [3] and identifying relevant research contributions. Researchers, students, and academicians are often required to analyze large numbers of research papers to understand existing methodologies, identify research gaps, compare approaches, and formulate new research directions. Manual analysis of scientific literature is time-consuming and can delay the research process.

To address these challenges, this project proposed and implemented a **Smart Research Paper Summarizer**, an AI-powered platform designed to automate research paper analysis and provide structured insights. The system integrates Natural Language Processing (NLP), Large Language Models (LLMs), PDF document processing, and intelligent information extraction techniques [1], [6], [7] to support researchers in understanding scientific literature efficiently.

The developed system goes beyond conventional summarization approaches by providing structured summaries, comparative analysis, survey synthesis, citation analysis, and figures-and-tables extraction through an interactive web-based interface.

5.2 Conclusion

This project successfully designed, developed, and evaluated a Smart Research Paper Summarizer capable of analyzing scientific documents and generating research-oriented outputs. The system was implemented using Python, Streamlit, PyMuPDF, Groq API, and the Llama 3.3 70B Large Language Model.

The proposed system addressed several limitations identified in existing research paper summarization tools. Unlike traditional summarizers that generate only generic summaries, the developed platform provides structured and comprehensive insights by extracting metadata, generating section-wise summaries, performing comparative analysis, synthesizing survey reports, analyzing citations, and extracting figures and tables from research papers.

The implementation consisted of five major modules:

1. Paper Analysis Module

The Paper Analysis module enables automatic extraction and summarization of individual research papers. It identifies important document sections and generates structured summaries that help users quickly understand the objectives, methodology, findings, contributions, and limitations of a research paper.

2. Comparative Analysis Module

The Comparative Analysis module supports side-by-side comparison of multiple research papers. This functionality assists researchers in evaluating different methodologies, identifying similarities and differences among studies, and recognizing emerging research trends.

3. Survey Synthesis Module

The Survey Synthesis module automatically generates literature review-style reports by consolidating information from multiple research papers. This feature significantly reduces the effort required to prepare survey papers and systematic literature reviews.

4. Citation Analysis Module

The Citation Analysis module extracts references and citation information from research papers, enabling users to review related work efficiently and identify influential studies within a research domain.

5. Figures and Tables Analysis Module

The Figures and Tables Analysis module extracts visual content such as figures, graphs, charts, tables, and captions. This feature allows users to quickly access important visual information without manually searching through lengthy documents.

The experimental evaluation demonstrated the effectiveness of the proposed system. The structured summarization approach achieved improved performance compared to conventional summarization techniques, particularly in terms of content coverage, readability, and usefulness for literature review tasks.

The obtained results showed that the proposed system achieved:

- Improved ROUGE scores compared to baseline methods.
- Higher usefulness ratings during human evaluation.
- Better support for literature review workflows.
- Effective comparative analysis capabilities.
- Enhanced understanding of research trends and contributions.

The integration of Large Language Models proved highly beneficial for generating coherent and context-aware summaries. Furthermore, the modular architecture of the system enables future expansion and integration of additional research assistance functionalities.

Overall, the developed Smart Research Paper Summarizer successfully demonstrates the feasibility of applying Artificial Intelligence and Large Language Models to automate research paper understanding and literature review processes. The system serves as an effective tool for researchers, students, and academicians by reducing the time and effort required to analyze scientific publications.

5.3 Major Contributions of the Project

The significant contributions of this project are summarized below:

Contribution 1: Structured Research Paper Summarization

Developed an AI-powered system capable of generating structured summaries rather than generic document summaries.

Contribution 2: Multi-Paper Comparative Analysis

Implemented a comparative analysis module that automatically compares research objectives, methodologies, findings, and contributions across multiple papers.

Contribution 3: Automated Survey Synthesis

Developed a survey generation mechanism that synthesizes information from multiple research papers into a literature review-style report.

Contribution 4: Citation Analysis

Implemented automatic extraction and analysis of references and citations from research papers.

Contribution 5: Figures and Tables Extraction

Designed a visual content extraction module capable of identifying and presenting figures, tables, and captions from research documents.

Contribution 6: User-Friendly Interface

Developed an interactive web-based interface using Streamlit for efficient user interaction and report generation.

Contribution 7: AI-Powered Research Assistant

Demonstrated the practical application of Large Language Models in assisting research paper analysis and knowledge discovery.

5.4 Limitations of the Current System

Although the proposed system achieved its objectives successfully, certain limitations remain.

Limited Dataset Size

The evaluation was conducted using a relatively small collection of research papers. Larger datasets may provide more comprehensive validation.

English Language Dependency

The current implementation supports only English-language research papers and does not handle multilingual documents.

Dependency on External APIs

The system relies on the Groq API and Large Language Models for summarization and analysis tasks. Service availability and API limits may affect performance.

Long Document Processing Challenges

Extremely large research papers may require additional optimization techniques for efficient processing.

Domain Dependency

The evaluation focused primarily on Computer Science and Engineering research papers. Performance across other domains requires further investigation.

Figure Interpretation

Although figures and tables can be extracted, the current version does not perform detailed semantic interpretation of visual content.

5.5 Future Scope

The Smart Research Paper Summarizer provides a strong foundation for future enhancements and advanced research assistance capabilities. Several improvements can be incorporated in future versions of the system.

5.5.1 Retrieval-Augmented Generation (RAG)

Future versions can integrate Retrieval-Augmented Generation (RAG) architectures to improve factual accuracy and reduce hallucinations generated by Large Language Models [10].

Benefits include:

- Improved reliability
- Better citation grounding
- Context-aware responses

5.5.2 Semantic Research Paper Search

A semantic search engine can be incorporated to enable users to search research papers using natural language queries instead of keywords.

Example:

"Find papers related to transformer-based scientific document summarization published after 2023."

Such functionality would significantly improve research discovery.

5.5.3 Citation Network Visualization

Future work may include graphical visualization of citation networks.

Features may include:

- Citation relationships
- Author collaboration networks
- Research influence analysis
- Knowledge graph generation

This would provide deeper insights into research ecosystems.

5.5.4 Research Gap Recommendation Engine

An intelligent recommendation system can be developed to automatically identify unexplored research areas and suggest potential future research directions.

Benefits:

- Faster research topic selection
- Enhanced innovation opportunities
- Better research planning

5.5.5 Multi-Language Support

Future versions can support research papers written in multiple languages such as:

- English
- French
- German
- Chinese
- Japanese

This would expand the applicability of the system to global research communities.

5.5.6 Scientific Knowledge Graph Construction

Research papers can be transformed into interconnected knowledge graphs representing:

- Concepts
- Methods
- Datasets
- Authors
- Citations

Knowledge graphs can improve information retrieval and scientific discovery.

5.5.7 Domain-Specific Fine-Tuned Models

Future implementations may utilize fine-tuned LLMs specialized for domains such as:

- Healthcare
- Artificial Intelligence
- Cybersecurity
- Mechanical Engineering
- Biotechnology

This can improve summary quality and domain understanding.

5.5.8 Cloud-Based Deployment

The current prototype can be extended into a scalable cloud platform capable of processing thousands of research papers.

Potential deployment platforms include:

- AWS
- Microsoft Azure
- Google Cloud Platform

Cloud deployment would improve accessibility and scalability.

5.5.9 Real-Time Research Monitoring

Future systems may continuously monitor newly published papers from sources such as:

- arXiv
- IEEE Xplore
- Springer
- ACM Digital Library
- PubMed

Researchers could receive automated updates and summaries relevant to their interests.

5.5.10 Intelligent Research Assistant

The ultimate future vision is the development of a complete AI Research Assistant capable of:

- Reading papers automatically
- Generating literature reviews
- Identifying research gaps
- Recommending methodologies
- Assisting with manuscript writing
- Supporting research planning

Such a system could transform the way scientific research is conducted.

5.6 Final Remarks

The Smart Research Paper Summarizer successfully demonstrates the potential of Artificial Intelligence, Natural Language Processing, and Large Language Models in addressing the challenges associated with scientific literature analysis. By integrating structured summarization, comparative analysis, survey synthesis, citation extraction, and visual content analysis within a single platform, the system significantly improves research productivity and accessibility.

The results obtained during implementation and evaluation confirm that AI-powered research assistance tools can effectively reduce the effort required for literature review and research exploration [2], [5], [20]. With further enhancements and integration of advanced AI technologies, the proposed system can evolve into a comprehensive intelligent research companion capable of supporting researchers throughout the entire research lifecycle.

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APPENDIX A

FULL SOURCE CODE

A.1 Introduction

The complete source code of the Smart Research Paper Summarizer is appended in this section for reference and verification of implementation. The source code demonstrates the development of a research paper analysis platform capable of metadata extraction, section detection, AI-powered summarization, comparative analysis, survey synthesis, citation extraction, and figures and tables analysis.

The application was developed using Python, Streamlit, PyMuPDF, Pandas, Plotly, and Groq API with the Llama 3.3 70B Versatile model.

The source code has been organized into the following modules:

1. Application Entry Point (app.py)
2. PDF Processing Module
3. Metadata Extraction Module
4. Section Detection Module
5. Text Chunking Module
6. Groq API Integration Module
7. Research Paper Summarization Module
8. Comparative Analysis Module
9. Survey Synthesis Module
10. Citation Analysis Module
11. Figures and Tables Extraction Module
12. User Interface Module
13. Experimental Evaluation Module
14. Error Handling and Validation Module

A.2 Application Entry Point (app.py)

Code: Application Entry and Configuration

```
```python
```

```
app.py (lines 1-57)
```

```
import base64
```

```
import logging
```

```
import os
```

```
import tempfile
```

```
from html import escape
```

```

import streamlit as st

from pipeline import DocumentExtractor, LLMService, crop_figure

st.set_page_config(page_title="Research Paper Summarizer", layout="wide")

DEFAULT_LLAMA_MODEL = os.getenv("OLLAMA_MODEL", "llama3.2:3b")

APP_VERSION = "2.0.2"

services = {}

def resolve_runtime_config(model_name: str) -> dict:

 model = model_name.strip() or DEFAULT_LLAMA_MODEL

 groq_key = os.getenv("GROQ_API_KEY", "").strip()

 if groq_key:

 runtime_provider = "groq"

 elif os.getenv("OLLAMA_BASE_URL", "").strip():

 runtime_provider = "ollama"

 else:

 runtime_provider = "groq"

 return {"provider": runtime_provider, "model": model}

def llm_config_signature(config: dict) -> tuple:

 return (

 config.get("provider", ""),

 config.get("model", ""),

 os.getenv("GROQ_API_KEY", "").strip(),

 os.getenv("OLLAMA_BASE_URL", "").strip(),

 os.getenv("SUMMARIX_MODEL_PATH", ""),

 os.getenv("SUMMARIX_REQUIRE_LLM", "").strip().lower(),

 APP_VERSION,

)

```

```

@st.cache_resource
def get_services(config_signature: tuple):
 provider = config_signature[0]

 llm_kwargs = {"provider": provider, "require_llm": False}

 if provider == "ollama":
 llm_kwargs["ollama_model"] = config_signature[1]

 return {
 "extractor": DocumentExtractor(),
 "llm": LLMService(**llm_kwargs),
 }
...

```

# Code: Main Application Loop and Tab Navigation

```

```python
# app.py (lines 472-609)

def main():
    global services

    with st.sidebar:
        st.markdown("### Research Paper Summarizer")
        st.markdown("Upload PDFs to analyze, compare, and summarize.")
        is_dark = True
        render_styles(is_dark)
        st.markdown("---")
        model_options = [
            "llama-3.3-70b-versatile",
            "llama-3.1-8b-instant",

```

```

        "gemma2-9b-it",
        "mixtral-8x7b-32768",
    ]
    model_name = st.selectbox(
        "LLM Model",
        options=model_options,
        index=0,
        help="Select the Groq-hosted model for summarization.",
    )
    runtime_config = resolve_runtime_config(model_name)
    runtime_provider = runtime_config["provider"]
    uploaded_files = st.file_uploader(
        "Upload arXiv Papers (PDF)",
        type=["pdf"],
        accept_multiple_files=True,
    )
    st.markdown("---")
    st.markdown(f"Runtime Backend: `{runtime_provider}`")
    os.environ["GROQ_MODEL"] = runtime_config.get("model", "llama-3.3-70b-versatile")
    config_signature = llm_config_signature(runtime_config)
    try:
        services = get_services(config_signature)
    except Exception as exc:
        st.error(llm_init_help(runtime_provider))
        st.exception(exc)
    return

```

```

llm = services.get("llm")

if llm and getattr(llm, "mode", "") == "fallback":

    st.warning(llm_init_help(runtime_provider), icon="warning")

    return

if not uploaded_files:

    st.info("Upload one or more PDFs from the sidebar to begin.")

    return

current_names = sorted(f.name for f in uploaded_files)

prev_names = st.session_state.get("uploaded_names", [])

files_changed = prev_names != current_names

if files_changed:

    st.session_state["uploaded_names"] = current_names

    old_papers = st.session_state.get("papers", {})

    st.session_state["papers"] = {k: v for k, v in old_papers.items() if k in current_names}

with st.spinner(f"Parsing {len(uploaded_files)} paper(s)..."):

    papers = parse_uploaded_files(uploaded_files)

tab_labels = ["Paper Analysis", "Comparative", "Survey", "Citations", "Figures & Tables"]

t1, t2, t3, t4, t5 = st.tabs(tab_labels)

with t1:

    tab_paper_analysis(papers)

with t2:

    tab_comparative(papers)

with t3:

    tab_survey(papers)

with t4:

    tab_citations(papers)

```

```

with t5:

    tab_figures_tables(papers)

if __name__ == "__main__":

    main()

...

```

A.3 PDF Processing Module

Code: PDF Text Extraction and Line Item Processing

```

```python

pipeline.py (lines 53-132)

import pymupdf as fitz

def _normalize_text(text: str) -> str:

 return re.sub(r"\s+", " ", text).strip()

class DocumentExtractor:

 def __init__(self, chunk_size: int = 420, chunk_overlap: int = 80):

 self.chunk_size = chunk_size

 self.chunk_overlap = chunk_overlap

 def parse_document(self, pdf_path: str, include_media: bool = True) -> Dict:

 if not os.path.exists(pdf_path):

 raise FileNotFoundError(pdf_path)

 lines = self._extract_line_items(pdf_path)

 metadata = self._extract_metadata(pdf_path, lines)

 sections = self._extract_sections(lines, metadata)

 figures: List[Dict] = []

 tables: List[Dict] = []

 if include_media:

 figures = self._extract_figures(pdf_path)

```

```

 tables = self._extract_tables(pdf_path)

 self._attach_media_to_sections(sections, figures, tables)

 citations = self.extract_citations(sections)

 return {

 "metadata": metadata,

 "sections": sections,

 "figures": figures,

 "tables": tables,

 "citations": citations,

 "pdf_path": pdf_path,

 }

def _extract_line_items(self, pdf_path: str) -> List[Dict]:

 items: List[Dict] = []

 with fitz.open(pdf_path) as doc:

 for page in doc:

 page_number = page.number + 1

 page_height = page.rect.height

 blocks = page.get_text("dict").get("blocks", [])

 for block in blocks:

 for line in block.get("lines", []):

 spans = line.get("spans", [])

 if not spans:

 continue

 text = "".join(span.get("text", "") for span in spans).strip()

 if not text:

 continue

```



```

y0 = min(span.get("bbox", [0, 0, 0, 0])[1] for span in spans)
y1 = max(span.get("bbox", [0, 0, 0, 0])[3] for span in spans)

if y0 < 24 or y1 > (page_height - 24):
 continue

item = {
 "page": page_number,
 "text": text,
 "size": max(span.get("size", 0) for span in spans),
 "font": " ".join(span.get("font", "") for span in spans).lower(),
 "y0": y0,
 "y1": y1,
 "page_height": page_height,
}

items.append(item)

return self._remove_repeating_headers_footers(items)

@staticmethod
def _remove_repeating_headers_footers(items: List[Dict]) -> List[Dict]:
 if not items:
 return items

 text_to_pages: Dict[str, set] = {}
 text_to_y: Dict[str, List[float]] = {}

 for item in items:
 text = _normalize_text(item["text"])

 if len(text) > 40 or len(text.split()) > 6:
 continue

```

```

 text_to_pages.setdefault(text, set()).add(item["page"])

 text_to_y.setdefault(text, []).append(item["y0"])

removable = set()

for text, pages in text_to_pages.items():

 if len(pages) < 3:

 continue

 y_med = median(text_to_y[text])

 if y_med < 95:

 removable.add(text)

filtered = []

for item in items:

 text = _normalize_text(item["text"])

 if text in removable:

 continue

 filtered.append(item)

return filtered
...

Code: Figure Cropping Utility

```python

# pipeline.py (lines 1104-1117)

def crop_figure(pdf_path: str, coords: Dict[str, float], output_path: str):

    with fitz.open(pdf_path) as doc:

        page_idx = int(coords["page"]) - 1

        if page_idx < 0 or page_idx >= len(doc):

            return

        page = doc[page_idx]

```

```

rect = fitz.Rect(
    coords["x"],
    coords["y"],
    coords["x"] + coords["w"],
    coords["y"] + coords["h"],
)

pix = page.get_pixmap(clip=rect, matrix=fitz.Matrix(2, 2))

pix.save(output_path)
...

```

A.4 Metadata Extraction Module

Code: Metadata Extraction

```

```python
pipeline.py (lines 164-289)

def _extract_metadata(self, pdf_path: str, lines: List[Dict]) -> Dict:
 with fitz.open(pdf_path) as doc:
 raw_meta = doc.metadata or {}

 title = self._safe_meta_value(raw_meta.get("title"))
 authors = self._safe_meta_value(raw_meta.get("author"))
 year = self._extract_year_from_meta(raw_meta)

 if not title:
 title = self._infer_title_from_text(lines)

 if not authors:
 authors = self._infer_authors_from_text(lines, title)

 if not year:
 year = self._infer_year_from_text(lines)

 return {

```

```

 "title": title or os.path.splitext(os.path.basename(pdf_path))[0],
 "authors": authors or "Unknown Authors",
 "year": year,
 }

 @staticmethod
 def _safe_meta_value(value: Optional[str]) -> Optional[str]:
 if not value:
 return None

 cleaned = _normalize_text(value)

 invalid_tokens = {
 "unknown", "untitled", "latex", "hyperref",
 "microsoft word", "adobe",
 }

 low = cleaned.lower()

 if any(token in low for token in invalid_tokens):
 return None

 if len(cleaned) <= 2:
 return None

 return cleaned

 def _infer_title_from_text(self, lines: List[Dict]) -> Optional[str]:
 first_page = [line for line in lines if line["page"] == 1]

 if not first_page:
 return None

 top_lines = [line for line in first_page if line["y0"] < 240]

 if not top_lines:
 return None

```

```

max_size = max(line["size"] for line in top_lines)

title_lines = [
 line for line in top_lines
 if line["size"] >= (max_size - 0.6)
 and len(line["text"].split()) >= 3
 and not self._looks_like_heading(line["text"])
 and "@" not in line["text"]
]

if not title_lines:
 candidates = sorted(top_lines, key=lambda x: (-x["size"], x["y0"]))
 return _normalize_text(candidates[0]["text"]) if candidates else None

title_lines = sorted(title_lines, key=lambda x: x["y0"])
title = " ".join(line["text"] for line in title_lines[:2])

return _normalize_text(title)

@staticmethod
def _infer_year_from_text(lines: List[Dict]) -> Optional[str]:
 years = re.findall(r"\b((?:19|20)\d{2})\b", " ".join(line["text"] for line in lines[:240]))

 if not years:
 return None

 return sorted(years)[-1]
...

```

## A.5 Section Detection Module

# Code: Section Detection and Heading Classification

```
```python
```

pipeline.py (lines 25-50, 291-402)

```
SECTION_HEADING_REGEX = re.compile(
```

```

r"^((\d+(\.\d+)*)|([IVXLCM]+))(\.|\s-)+[A-Z][A-Za-z0-9\-\s,():/]+$"
)
COMMON_SECTION_TITLES = {
    "abstract", "introduction", "background", "related work",
    "method", "methods", "methodology", "approach", "model",
    "experiments", "experimental setup", "results", "discussion",
    "conclusion", "conclusions", "future work", "limitations",
    "appendix", "references", "acknowledgments",
}

def _extract_sections(self, lines: List[Dict], metadata: Dict) -> Dict[str, Dict]:
    if not lines:
        return {}

    body_sizes = [line["size"] for line in lines if len(line["text"].split()) >= 4]
    body_size = median(body_sizes) if body_sizes else 10.5
    heading_positions = []

    for idx, line in enumerate(lines):
        if self._is_section_heading(line, body_size, metadata):
            heading_positions.append(idx)

    anchor_idx = self._find_heading_anchor(heading_positions, lines)

    if anchor_idx is not None:
        heading_positions = [idx for idx in heading_positions if idx >= anchor_idx]

    if not heading_positions:
        joined = "\n".join(line["text"] for line in lines)
        return {
            "Document Content": self._build_section(
                "Document Content", joined, {line["page"] for line in lines}

```

```

    )
}

sections: Dict[str, Dict] = {}

for pos, start_idx in enumerate(heading_positions):
    end_idx = heading_positions[pos + 1] \
        if pos + 1 < len(heading_positions) else len(lines)
    heading_line = lines[start_idx]
    title = _normalize_text(heading_line["text"].rstrip(":"))
    body_lines = lines[start_idx + 1: end_idx]
    content_lines = [line["text"] for line in body_lines]
    pages = {heading_line["page"]}
    pages.update(line["page"] for line in body_lines)
    key = self._deduplicate_title(sections, title)
    sections[key] = self._build_section(key, "\n".join(content_lines), pages)

return sections

def _is_section_heading(self, line: Dict, body_size: float, metadata: Dict) -> bool:
    text = _normalize_text(line["text"])
    if len(text) < 2 or len(text) > 120:
        return False
    if text.lower() == metadata.get("title", "").lower():
        return False
    if text.lower().startswith(("figure ", "fig.", "table ", "algorithm ")):
        return False
    if text.lower().startswith("arxiv."):
        return False
    if "@" in text:

```

```

        return False

    lower = text.lower()

    words = text.split()

    if line["page"] == 1 and line["y0"] < 360:

        if lower not in {"abstract", "introduction"} \

            and not SECTION_HEADING_REGEX.match(text):

            return False

    if lower in COMMON_SECTION_TITLES and line["size"] >= body_size:

        return True

    if SECTION_HEADING_REGEX.match(text) and len(words) <= 14:

        return True

    looks_bold = "bold" in line["font"] or "semibold" in line["font"]

    if len(words) <= 12 and text[-1] not in ".,:":

        if line["size"] >= body_size + 1.0:

            return True

        if looks_bold and line["size"] >= body_size + 0.3 \

            and self._looks_like_title_case(text):

            return True

    return False

'''

```

A.6 Text Chunking Module

Code: Text Chunking

```
'''python
```

pipeline.py (lines 57-67)

```
def chunk_text(text: str, max_words: int = 450, overlap: int = 80) -> List[str]:
```

```
    words = text.split()
```



```

if not words:

    return []

stride = max(max_words - overlap, 1)

chunks = []

start = 0

while start < len(words):

    chunks.append(" ".join(words[start: start + max_words]))

    start += stride

return chunks
...

# Code: Section Builder with Chunking
```python
pipeline.py (lines 411-423)

def _build_section(self, title: str, content: str, pages: set) -> Dict:

 normalized_content = content.strip()

 chunks = chunk_text(normalized_content, self.chunk_size, self.chunk_overlap)

 return {

 "title": title,

 "content": normalized_content,

 "pages": sorted(pages),

 "chunks": chunks,

 "summary": "",

 "chunk_count": len(chunks),

 "figures": [],

 "tables": [],

 }

```

...

## A.7 Groq API Integration Module

# Code: Groq Client Initialization

```python

pipeline.py (lines 556-650)

class LLMService:

def __init__(

self,

model_path: Optional[str] = None,

provider: Optional[str] = None,

require_llm: Optional[bool] = None,

ollama_model: Optional[str] = None,

):

self.llm = None

self.model_path = model_path or os.getenv(

"SUMMARIX_MODEL_PATH", "models/llama-3.2-1b-instruct.Q4_K_M.gguf"

)

self.provider = (provider or os.getenv("SUMMARIX_LLM_PROVIDER",
 "")).strip().lower()

self.require_llm = self._env_flag("SUMMARIX_REQUIRE_LLM", default=False) \

if require_llm is None else bool(require_llm)

self.mode = "fallback"

self.groq_api_key = os.getenv("GROQ_API_KEY", "").strip()

self.groq_model = os.getenv("GROQ_MODEL", "llama-3.3-70b-versatile")

self.groq_max_input_chars = self._env_int("GROQ_MAX_INPUT_CHARS",
 default=12000, minimum=2000)

self.groq_max_retries = self._env_int("GROQ_MAX_RETRIES", default=3,
 minimum=1)

```

self._groq_client = None

if not self.provider:

    if self.groq_api_key:

        self.provider = "groq"

    elif self.ollama_base_url:

        self.provider = "ollama"

    else:

        self.provider = "local"

init_error: Optional[Exception] = None

try:

    if self.provider == "groq":

        self._init_groq()

    elif self.provider == "ollama":

        self._init_ollama()

    elif self.provider == "local":

        self._init_local()

except Exception as exc:

    init_error = exc

if init_error is not None:

    if self.require_llm:

        raise RuntimeError(

            "LLM initialization failed. Set GROQ_API_KEY for Groq, "

            "or OLLAMA_BASE_URL + OLLAMA_MODEL for Ollama, "

            "or provide a valid local GGUF model via SUMMARIX_MODEL_PATH."

        ) from init_error

    logging.warning("LLM unavailable, fallback summarizer enabled: %s", init_error)

```

```

        self.mode = "fallback"

def _init_groq(self):
    if not self.groq_api_key:
        raise RuntimeError("GROQ_API_KEY is missing for provider='groq'.")

    try:
        from groq import Groq

        self._groq_client = Groq(api_key=self.groq_api_key)

        self._groq_client.models.list()

    except ImportError as exc:
        raise RuntimeError(
            "groq package is not installed. Add 'groq>=0.11,<1' to requirements.txt."
        ) from exc

    except Exception as exc:
        raise RuntimeError(f"Groq API initialization failed: {exc}") from exc

    self.mode = "groq"

...

# Code: Groq Summarization Request

```python
pipeline.py (lines 734-790)

def _summarize_groq(self, text: str, context: str = "") -> str:
 if self._groq_client is None:
 raise RuntimeError("Groq client is not initialized.")

 max_chars_seq = [
 self.groq_max_input_chars,
 int(self.groq_max_input_chars * 0.75),
 int(self.groq_max_input_chars * 0.5),

```

```

]

last_error: Optional[Exception] = None

for max_chars in max_chars_seq:

 clipped_text = self._truncate_for_context(text, max_chars)

 messages = [

 {

 "role": "system",

 "content": "You are an expert research assistant. "

 "Write concise, accurate summaries of research paper sections.",

 },

 {

 "role": "user",

 "content": (

 f"Context: {context}\n"

 f"Section text:\n{clipped_text}\n\n"

 "Write a concise summary of the above section:"

),

 },

]

for attempt in range(1, self.groq_max_retries + 1):

 try:

 response = self._groq_client.chat.completions.create(

 model=self.groq_model,

 messages=messages,

 max_tokens=300,

 temperature=0.2,

```

```

)

 generated = response.choices[0].message.content.strip()

 if generated:

 return generated

except Exception as exc:

 last_error = exc

 msg = str(exc).lower()

 if "context" in msg and ("length" in msg or "window" in msg or "token" in msg):

 break

 if "rate" in msg and attempt < self.groq_max_retries:

 time.sleep(min(2.0 * attempt, 8.0))

 continue

 if self.require_llm:

 raise RuntimeError(f"Groq summarization failed: {exc}") from exc

 logging.warning("Groq summarization failed, using fallback: %s", exc)

 return self._fallback_summary(text)

if self.require_llm and last_error:

 raise RuntimeError(f"Groq summarization failed after retries: {last_error}")

 logging.warning("Groq summarization failed after retries, using fallback summary")

 return self._fallback_summary(text)
...

Code: Generic LLM Generation Dispatch

```python

# pipeline.py (lines 1051-1085)

def _llm_generate(self, prompt: str, max_tokens: int = 600) -> str:

    if self.mode == "groq":

```

```

        return self._groq_generate(prompt, max_tokens)

    if self.mode == "ollama":

        return self._ollama_generate_text(prompt, max_tokens)

    if self.mode == "local-llama":

        return self._local_generate(prompt, max_tokens)

    return self._fallback_summary(prompt)

def _groq_generate(self, prompt: str, max_tokens: int = 600) -> str:

    if self._groq_client is None:

        raise RuntimeError("Groq client is not initialized.")

    messages = [

        {"role": "system", "content": "You are an expert research analyst."},

        {"role": "user", "content": prompt},

    ]

    for attempt in range(1, self.groq_max_retries + 1):

        try:

            response = self._groq_client.chat.completions.create(

                model=self.groq_model,

                messages=messages,

                max_tokens=max_tokens,

                temperature=0.3,

            )

            generated = response.choices[0].message.content.strip()

            if generated:

                return generated

        except Exception as exc:

            msg = str(exc).lower()

```

```

        if "rate" in msg and attempt < self.groq_max_retries:

            time.sleep(min(2.0 * attempt, 8.0))

            continue

        raise RuntimeError(f"Groq generation failed: {exc}") from exc

    return "Generation failed after retries."
...

```

A.8 Research Paper Summarization Module

Code: Core Summarization Dispatch

```

```python
pipeline.py (lines 713-732)

def summarize(self, text: str, context: str = "") -> str:

 if not text.strip():

 return "No content found for summarization."

 if self.mode == "groq":

 return self._summarize_groq(text, context)

 if self.mode == "ollama":

 return self._summarize_ollama(text, context)

 if self.mode == "local-llama":

 return self._summarize_local(text, context)

 if self.require_llm:

 raise RuntimeError(

 "LLM is required but unavailable. Set GROQ_API_KEY, "

 "OLLAMA_BASE_URL + OLLAMA_MODEL, "

 "or local GGUF via SUMMARIX_MODEL_PATH."

)

 return self._fallback_summary(text)

```



```

...

Code: Section Summary with Caching (app.py)

```python
# app.py (lines 159-181)

def section_summary(title: str, section: dict, metadata: dict) -> str:

    cache = st.session_state.setdefault("section_summaries", {})

    doc_key = f"{metadata.get('title', '')}::{title}"

    if doc_key in cache:

        return cache[doc_key]

    text = " ".join(section.get("chunks", [])[:3]) or section.get("content", "")

    context = f"{metadata.get('title', 'Paper')} -> {title}"

    try:

        summary = services["llm"].summarize(text, context)

    except Exception as exc:

        msg = str(exc).strip() or "Unknown LLM failure."

        summary = f"LLM error: {msg}"

    cache[doc_key] = summary

    return summary

def summarize_all_sections(sections: dict, metadata: dict):

    for title, section in sections.items():

        section_summary(title, section, metadata)
...

```

A.9 Comparative Analysis Module

```

# Code: Comparative Analysis Generation

```python

```

```

pipeline.py (lines 975-1014)

def compare_papers(self, papers: List[Dict]) -> str:

 if not papers or len(papers) < 2:

 return "At least two papers are required for comparative analysis."

 paper_briefs = []

 for i, paper in enumerate(papers, 1):

 meta = paper.get("metadata", {})

 sections = paper.get("sections", {})

 abstract = ""

 conclusion = ""

 for title, sec in sections.items():

 low = title.lower().strip()

 if low == "abstract" and not abstract:

 abstract = self._truncate_for_context(sec.get("content", ""), 800)

 if low in {"conclusion", "conclusions"} and not conclusion:

 conclusion = self._truncate_for_context(sec.get("content", ""), 600)

 brief = (

 f"Paper {i}: {meta.get('title', 'Untitled')}\n"

 f"Authors: {meta.get('authors', 'Unknown')}\n"

 f"Year: {meta.get('year', 'N/A')}\n"

 f"Abstract: {abstract or 'Not available'}\n"

 f"Conclusion: {conclusion or 'Not available'}\n"

)

 paper_briefs.append(brief)

 combined = "\n---\n".join(paper_briefs)

 prompt_text = (

```

```

f"Below are summaries of {len(papers)} research papers:\n\n"

f"{combined}\n\n"

"Provide a structured comparative analysis covering:\n"

"1. Research Objectives\n"

"2. Methodologies\n"

"3. Key Findings\n"

"4. Datasets & Evaluation\n"

"5. Strengths & Limitations\n"

"6. Agreements & Contradictions\n\n"

"Format with clear headings and bullet points."

)

return self._llm_generate(prompt_text, max_tokens=1200)
...

Code: Comparative Tab UI (app.py)

```python
# app.py (lines 319-355)

def tab_comparative(papers: dict):

    names = list(papers.keys())

    if len(names) < 2:

        st.info("Upload at least 2 papers to enable comparative analysis.")

    return

    st.subheader("Paper Overview")

    header="<tr><th>#</th><th>Title</th><th>Authors</th><th>Year</th><th>Sections</th><th>
Citations</th></tr>"

    rows = ""

    for i, (name, data) in enumerate(papers.items(), 1):

        m = data.get("metadata", {})

```

```

rows += (

    f"<tr><td>{i}</td>"

    f"<td>{escape(m.get('title', 'Untitled'))}</td>"

    f"<td>{escape(m.get('authors', 'Unknown'))}</td>"

    f"<td>{escape(str(m.get('year', 'N/A')))}</td>"

    f"<td>{len(data.get('sections', {}))}</td>"

    f"<td>{len(data.get('citations', []))}</td></tr>"

)

st.markdown(f"<table class='cmp-table'>{header}{rows}</table>",
            unsafe_allow_html=True)

st.markdown("---")

st.subheader("AI Comparative Analysis")

cache_key = "comparative_result"

if st.button("Generate Comparative Analysis", key="cmp_btn",
use_container_width=True):

    with st.spinner("Analyzing papers..."):

        try:

            result = services["llm"].compare_papers(list(papers.values()))

            st.session_state[cache_key] = result

        except Exception as exc:

            st.error(f"Comparison failed: {exc}")

    result = st.session_state.get(cache_key, "")

    if result:

        st.markdown(result)

...

```

A.10 Survey Synthesis Module

Code: Survey Synthesis Generation

```

```python
pipeline.py (lines 1016-1049)

def synthesize_survey(self, papers: List[Dict]) -> str:

 if not papers:

 return "No papers provided for survey synthesis."

 paper_briefs = []

 for i, paper in enumerate(papers, 1):

 meta = paper.get("metadata", {})

 sections = paper.get("sections", {})

 abstract = ""

 for title, sec in sections.items():

 if title.lower().strip() == "abstract":

 abstract = self._truncate_for_context(sec.get("content", ""), 800)

 break

 brief = (

 f"Paper {i}: {meta.get('title', 'Untitled')} "

 f"({meta.get('authors', 'Unknown')}, {meta.get('year', 'N/A')})\n"

 f"Abstract: {abstract or 'Not available'}"

)

 paper_briefs.append(brief)

 combined = "\n---\n".join(paper_briefs)

 prompt_text = (

 f"Below are abstracts from {len(papers)} research papers on related topics:\n\n"

 f"{combined}\n\n"

 "Write a unified survey-style summary that:\n"

 "1. Identifies the **overarching research theme**\n"

```

```

"2. Groups papers by **sub-themes or approaches**\n"
"3. Traces the **evolution of ideas** across papers\n"
"4. Highlights **open problems and future directions**\n"
"5. Provides **a concluding synthesis** of the field's state\n\n"
"Write in a formal academic tone with clear paragraphs."
)

return self._llm_generate(prompt_text, max_tokens=1500)
...

Code: Survey Tab UI (app.py)
```python
# app.py (lines 360-380)
def tab_survey(papers: dict):
    names = list(papers.keys())

    if not names:
        st.info("Upload papers first.")
        return

    st.subheader("AI Survey Synthesis")

    st.caption(f"Synthesizing a unified survey across {len(names)} paper(s).")

    cache_key = "survey_result"

    if st.button("Generate Survey Summary", key="survey_btn", use_container_width=True):
        with st.spinner("Generating thematic survey synthesis..."):
            try:
                result = services["llm"].synthesize_survey(list(papers.values()))

                st.session_state[cache_key] = result

            except Exception as exc:
                st.error(f"Survey synthesis failed: {exc}")

```

```

result = st.session_state.get(cache_key, "")

if result:

    st.markdown(result)

...

```

A.11 Citation Analysis Module

Code: Citation Extraction

```

```python

pipeline.py (lines 503-553)

@staticmethod

def extract_citations(sections: Dict[str, Dict]) -> List[Dict]:

 ref_section = None

 for title, sec in sections.items():

 if title.lower().strip() in {"references", "bibliography", "works cited"}:

 ref_section = sec

 break

 if ref_section is None:

 return []

 raw = ref_section.get("content", "")

 if not raw.strip():

 return []

 citations: List[Dict] = []

 # Strategy 1: numbered references [1], [2], ... or 1. 2. ...

 numbered = re.split(r"\n?[?(\d{1,3})\]?\.?\s+", raw)

 if len(numbered) >= 5:

 idx = 1

 while idx + 1 < len(numbered):

```

```

num = numbered[idx].strip()

text = _normalize_text(numbered[idx + 1])

if len(text) > 15:

 citations.append({

 "id": f"ref_{num}",

 "number": int(num),

 "text": text,

 "pages": ref_section.get("pages", []),

 })

 idx += 2

if citations:

 return citations

Strategy 2: split on double-newlines

blocks = re.split(r"\n{2,}", raw)

if len(blocks) < 3:

 blocks = re.split(r"\n(?:=[A-Z][a-z]+,?\s+[A-Z])", raw)

for i, block in enumerate(blocks, start=1):

 text = _normalize_text(block)

 if len(text) > 15:

 citations.append({

 "id": f"ref_{i}",

 "number": i,

 "text": text,

 "pages": ref_section.get("pages", []),

 })

return citations

```



```

...

Code: Citations Tab UI (app.py)

```python

# app.py (lines 385-409)

def tab_citations(papers: dict):

    if not papers:

        st.info("Upload papers to extract citations.")

        return

    for name, data in papers.items():

        citations = data.get("citations", [])

        meta = data.get("metadata", {})

        title = meta.get("title", name)

        with st.expander(f'{title} - {len(citations)} reference(s)',

                        expanded=len(papers) == 1):

            if not citations:

                st.caption("No citations could be extracted from this paper.")

                continue

            html_items = ""

            for cite in citations:

                num = cite.get("number", "?")

                text = escape(cite.get("text", ""))

                html_items += f'<div class="cite-item">' \

                             f'<span class="cite-num">[{num}]</span>{text}</div>'

            st.markdown(

                f'<div class="panel" style="max-height: 500px; overflow-y: auto;">'

                f'{html_items}</div>',

```

```

        unsafe_allow_html=True,
    )
'''

```

A.12 Figures and Tables Extraction Module

Code: Figure Extraction

```

'''python
# pipeline.py (lines 425-453)

def _extract_figures(self, pdf_path: str) -> List[Dict]:

    figures: List[Dict] = []

    with fitz.open(pdf_path) as doc:

        for page in doc:

            page_number = page.number + 1

            images = page.get_images(full=True)

            for img_index, image in enumerate(images, start=1):

                xref = image[0]

                rects = page.get_image_rects(xref)

                if not rects:

                    continue

                rect = rects[0]

                figures.append({

                    "id": f"figure_{page_number}_{img_index}",

                    "label": f"Figure {len(figures) + 1}",

                    "description": f"Image extracted from page {page_number}",

                    "page": page_number,

                    "coords": {

                        "page": page_number,

```

```

        "x": rect.x0,

        "y": rect.y0,

        "w": max(rect.width, 10),

        "h": max(rect.height, 10),

    },

    "summary": "",

})

return figures
...

# Code: Table Extraction

```python
pipeline.py (lines 455-494)

def _extract_tables(self, pdf_path: str) -> List[Dict]:

 tables: List[Dict] = []

 with fitz.open(pdf_path) as doc:

 for page in doc:

 page_number = page.number + 1

 if not hasattr(page, "find_tables"):

 continue

 try:

 found = page.find_tables()

 except Exception as exc:

 logging.warning("Table detection failed on page %s: %s",

 page_number, exc)

 continue

 for idx, table in enumerate(found.tables, start=1):

```

```

 rows = table.extract() if hasattr(table, "extract") else []

 preview = self._table_preview(rows)

 bbox = table.bbox if getattr(table, "bbox", None) else (0, 0, 0, 0)

 tables.append({
 "id": f"table_{page_number}_{idx}",
 "label": f"Table {len(tables) + 1}",
 "page": page_number,
 "bbox": bbox,
 "preview": preview,
 "summary": "",
 })

 return tables

@staticmethod
def _table_preview(rows: List) -> str:
 if not rows:
 return "Table detected, but cells could not be parsed."

 preview_lines = []

 for row in rows[:4]:
 if isinstance(row, dict):
 cells = list(row.values())
 else:
 cells = list(row)

 preview_lines.append(" | ".join(
 _normalize_text(str(cell)) for cell in cells))

 return "\n".join(preview_lines)

```

# Code: Media-to-Section Attachment

```
```python
```

pipeline.py (lines 496-501)

```
@staticmethod
```

```
def _attach_media_to_sections(sections: Dict[str, Dict],
```

```
    figures: List[Dict],
```

```
    tables: List[Dict]):
```

```
    for section in sections.values():
```

```
        pages = set(section.get("pages", []))
```

```
        section["figures"] = [fig["id"] for fig in figures if fig["page"] in pages]
```

```
        section["tables"] = [tab["id"] for tab in tables if tab["page"] in pages]
```

```
```
```

## A.13 User Interface Module

# Code: CSS Styling Engine

```
```python
```

app.py (lines 88-154)

```
def render_styles(is_dark: bool = False):
```

```
    st.markdown(
```

```
        f"""
```

```
        <style>
```

```
@import url("https://fonts.googleapis.com/css2?family=Public+Sans:wght@400;600;700&display=swap");
```

```
@import url("https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/css/all.min.css");
```

```
:root {{
```

```
    --bg-primary: {"#0f1117" if is_dark else "#ffffff"};
```

```
    --bg-secondary: {"#1a1d27" if is_dark else "#f8fbff"};
```

```
    --bg-card: {"#1e2130" if is_dark else "#ffffff"};
```

```

--bg-panel:    {"#252838" if is_dark else "#f4f8fe"};
--border:      {"#2d3148" if is_dark else "#dbe4f0"};
--text-primary: {"#e2e8f0" if is_dark else "#0f172a"};
--text-secondary: {"#94a3b8" if is_dark else "#334155"};
--text-muted:   {"#64748b" if is_dark else "#475569"};
--accent:      {"#60a5fa" if is_dark else "#3b82f6"};
}}

html, body, [class*="css"] {{ font-family: "Public Sans", sans-serif; }}

.hero {{
    border: 1px solid var(--border);
    background: var(--hero-bg);
    border-radius: 16px; padding: 20px; margin-bottom: 18px;
}}

.hero h1 {{ margin: 0; color: var(--text-primary); font-size: 1.8rem; }}
.hero p {{ margin: 8px 0 0 0; color: var(--text-secondary); }}

.meta-card {{
    border: 1px solid var(--border); border-radius: 14px;
    background: var(--bg-card); padding: 18px; margin-bottom: 16px;
}}

.meta-row {{ display: grid; grid-template-columns: 160px 1fr; gap: 10px;
    margin-bottom: 10px; color: var(--text-primary); }}

.meta-key {{ color: var(--text-muted); font-weight: 600; }}

.panel {{ border: 1px solid var(--border); border-radius: 12px;
    padding: 14px; background: var(--bg-card); }}

.panel-title {{ font-weight: 700; margin-bottom: 10px;
    color: var(--text-primary); }}

```

```

.source-box {{ height: 440px; overflow-y: auto; line-height: 1.65;
    color: var(--text-primary); font-size: 0.95rem;
    white-space: normal; }}

.summary-box {{ min-height: 220px; line-height: 1.7;
    color: var(--text-primary); background: var(--bg-panel);
    border: 1px solid var(--border); border-radius: 10px;
    padding: 12px; margin-top: 12px; }}

.cmp-table {{ width: 100%; border-collapse: collapse; margin-top: 12px; }}
.cmp-table th, .cmp-table td {{ border: 1px solid var(--border);
    padding: 10px 14px; text-align: left; font-size: 0.93rem; }}
.cmp-table th {{ background: var(--table-header); font-weight: 700;
    color: var(--text-primary); }}

.fig-gallery {{ display: grid; grid-template-columns: repeat(auto-fill, minmax(280px, 1fr));
    gap: 16px; margin-top: 12px; }}

.fig-card {{ border: 1px solid var(--border); border-radius: 12px;
    padding: 12px; background: var(--bg-card); text-align: center; }}

</style>

"""
    unsafe_allow_html=True,
)
...

# Code: Metadata Card and Section Viewer

```python
app.py (lines 211-294)

def render_metadata_card(metadata: dict, section_count: int,
 citation_count: int, fig_count: int, table_count: int):

```

```

title = escape(metadata.get("title") or "Untitled Document")

authors = escape(metadata.get("authors") or "Unknown Authors")

year = escape(str(metadata.get("year") or "N/A"))

st.markdown(
 f"""
 <div class="meta-card">

 <div class="meta-row"><div class="meta-key">Title</div><div>{title}</div></div>

 <div class="meta-row"><div
class="meta-key">Authors</div><div>{authors}</div></div>

 <div class="meta-row"><div class="meta-key">Year</div><div>{year}</div></div>

 <div class="meta-row"><div
class="meta-key">Sections</div><div>{section_count}</div></div>

 <div class="meta-row"><div
class="meta-key">Citations</div><div>{citation_count}</div></div>

 <div class="meta-row"><div class="meta-key">Figures / Tables</div>

 <div>{fig_count} / {table_count}</div></div>

 </div>

 """,
 unsafe_allow_html=True,
)

def render_sections(sections: dict, metadata: dict):
 titles = list(sections.keys())

 if not titles:
 st.warning("No sections detected in this document.")

 return

if st.button("Summarize All Sections",
 key=f"sumall_{metadata.get('title', '')}",
 use_container_width=True):

```



```

with st.spinner("Generating summaries for all sections"):

 summarize_all_sections(sections, metadata)

tabs = st.tabs(titles)

for title, tab in zip(titles, tabs):

 section = sections[title]

 with tab:

 pages = ", ".join(str(p) for p in section.get("pages", [])) or "N/A"

 chunk_count = section.get("chunk_count", 0)

 word_count = len(section.get("content", "").split())

 st.markdown(

 f"""<div class="section-info">

 Pages: {pages}

 Chunks: {chunk_count}

 Words: {word_count}

 </div>""",

 unsafe_allow_html=True,

)

 left, right = st.columns([1.2, 1])

 with left:

 source = escape(section.get("content") or "No text found.") \

 .replace("\n", "
")

 st.markdown(

 f"""<div class="panel">

 <div class="panel-title">Original Content</div>

 <div class="source-box">{source}</div>

 </div>""",

```

```

 unsafe_allow_html=True,

)

with right:

 btn_key = f"sum_{metadata.get('title', '')}_{title}"

 if st.button("Summarize This Section",

 key=btn_key, use_container_width=True):

 with st.spinner(f"Summarizing: {title}"):

 section_summary(title, section, metadata)

 doc_key = f"{metadata.get('title', '')}::{title}"

 summary = st.session_state.get("section_summaries", {}).get(doc_key, "")

 if summary:

 rendered = escape(summary)

 else:

 rendered = "Click Summarize This Section " \

 "to generate summary."

 st.markdown(

 f"""<div class="panel">

 <div class="panel-title">Section Summary</div>

 <div class="summary-box">{rendered}</div>

 </div>""",

 unsafe_allow_html=True,

)

...

```

## A.14 Experimental Evaluation Module

# Code: ROUGE Metrics Implementation

```
```python
```

```

# research_experiment_framework.py (lines 78-325)

from collections import Counter

import math

def tokenize(text: str) -> List[str]:
    return [t for t in re.findall(r"[A-Za-z0-9]+", text.lower())
            if t and t not in STOPWORDS]

def cosine_sim(counter_a: Counter, counter_b: Counter) -> float:
    if not counter_a or not counter_b:
        return 0.0

    keys = set(counter_a) | set(counter_b)

    dot = sum(counter_a.get(k, 0) * counter_b.get(k, 0) for k in keys)

    norm_a = math.sqrt(sum(v * v for v in counter_a.values()))
    norm_b = math.sqrt(sum(v * v for v in counter_b.values()))

    if norm_a == 0 or norm_b == 0:
        return 0.0

    return dot / (norm_a * norm_b)

def jaccard(tokens_a: List[str], tokens_b: List[str]) -> float:
    set_a, set_b = set(tokens_a), set(tokens_b)

    if not set_a or not set_b:
        return 0.0

    return len(set_a & set_b) / len(set_a | set_b)

class EvaluationFramework:

    @staticmethod
    def rouge_n_f1(candidate: str, reference: str, n: int = 1) -> float:
        def ngrams(tokens: List[str], nsize: int) -> List[Tuple[str, ...]]:
            return [tuple(tokens[i:i + nsize])
                    for i in range(len(tokens) - nsize + 1)]

```

```

        for i in range(max(0, len(tokens) - nsize + 1))]

c_tokens = tokenize(candidate)

r_tokens = tokenize(reference)

c_ngrams = Counter(ngrams(c_tokens, n))

r_ngrams = Counter(ngrams(r_tokens, n))

if not c_ngrams or not r_ngrams:

    return 0.0

overlap = sum((c_ngrams & r_ngrams).values())

precision = overlap / max(1, sum(c_ngrams.values()))

recall = overlap / max(1, sum(r_ngrams.values()))

if precision + recall == 0:

    return 0.0

return 2 * precision * recall / (precision + recall)

@staticmethod
def rouge_l_f1(candidate: str, reference: str) -> float:

    c = tokenize(candidate)

    r = tokenize(reference)

    if not c or not r:

        return 0.0

    dp = [[0] * (len(r) + 1) for _ in range(len(c) + 1)]

    for i in range(1, len(c) + 1):

        for j in range(1, len(r) + 1):

            if c[i - 1] == r[j - 1]:

                dp[i][j] = dp[i - 1][j - 1] + 1

            else:

                dp[i][j] = max(dp[i - 1][j], dp[i][j - 1])

```

```

lcs = dp[-1][-1]

precision = lcs / max(1, len(c))

recall = lcs / max(1, len(r))

if precision + recall == 0:

    return 0.0

return 2 * precision * recall / (precision + recall)

def evaluate_summary(self, candidate: str, reference: str,

                    source_text: str) -> Dict:

    audit = self.fact_checker.audit(candidate, source_text)

    return {

        "rouge1_f1": round(self.rouge_n_f1(candidate, reference, 1), 4),

        "rouge2_f1": round(self.rouge_n_f1(candidate, reference, 2), 4),

        "rougeL_f1": round(self.rouge_l_f1(candidate, reference), 4),

        "semantic_f1_proxy": round(

            self.semantic_f1_proxy(candidate, reference), 4),

        "factual_score": audit.factual_score,

        "contradictions": audit.contradiction_count,

    }

...

# Code: Factual Consistency Checker

```python

research_experiment_framework.py (lines 194-273)

class FactualConsistencyChecker:

 def __init__(self):

 self.neg_terms = {"no", "not", "never", "none", "without",

```

```

 "cannot", "can't", "won't"}

def audit(self, summary: str, source_text: str,

 threshold: float = 0.17) -> SummaryAudit:

 source_sentences = split_sentences(source_text)

 sum_sentences = split_sentences(summary)

 if not sum_sentences:

 return SummaryAudit(summary=summary, factual_score=0.0,

 contradiction_count=0, flagged_sentences=[])

 supports = []

 flagged = []

 contradiction_count = 0

 for s in sum_sentences:

 best_src, score = self._best_support_sentence(s, source_sentences)

 contradiction = self._is_contradiction(s, best_src)

 if contradiction:

 contradiction_count += 1

 supports.append(score)

 if score < threshold or contradiction:

 flagged.append({

 "summary_sentence": s,

 "best_source_sentence": best_src,

 "support_score": round(score, 4),

 "contradiction": contradiction,

 })

 base = sum(supports) / len(supports)

 penalty = contradiction_count * 0.08

```

```

factual_score = max(0.0, min(1.0, base - penalty))

return SummaryAudit(summary=summary,

 factual_score=round(factual_score, 4),

 contradiction_count=contradiction_count,

 flagged_sentences=flagged)

def _is_contradiction(self, summary_sentence: str,

 support_sentence: str) -> bool:

 sum_low = summary_sentence.lower()

 src_low = support_sentence.lower()

 sum_neg = any(n in sum_low for n in self.neg_terms)

 src_neg = any(n in src_low for n in self.neg_terms)

 if sum_neg != src_neg and support_sentence:

 return True

 sum_nums = set(self._numbers(summary_sentence))

 src_nums = set(self._numbers(support_sentence))

 if sum_nums and src_nums and not (sum_nums & src_nums):

 return True

 return False

...

Code: Experiment Runner

```python

# run_research_experiments.py (lines 1-40)

import json

from pathlib import Path

from research_experiment_framework import ResearchPipeline, pretty_metric_table

def main():

```

```

root = Path(__file__).resolve().parent
pdf_paths = [str(root / "data" / "2004.05150v2.pdf")]
pipeline = ResearchPipeline(use_local_model=True)
if not pipeline.llm.has_local_model():
    raise RuntimeError("Local llama.cpp model failed to load")
papers = pipeline.load_papers(pdf_paths, include_media=True)
single = pipeline.run_single_paper_experiment(papers[0])
phase2_media = pipeline.run_phase2_media_experiment(papers[0])
multi = pipeline.run_multi_document_experiment(
    papers if len(papers) > 1 else papers * 3
)
payload = {
    "single": single,
    "phase2_media": phase2_media,
    "multi": multi,
    "metric_table": pretty_metric_table(single),
}
out_file = root / "research_experiment_results.json"
out_file.write_text(json.dumps(payload, indent=2), encoding="utf-8")
print("Saved:", out_file)
print(payload["metric_table"])
if __name__ == "__main__":
    main()
...

```

A.15 Error Handling and Validation

Code: API Error Handling and Retry Logic


```

```python
pipeline.py (lines 760-784, 1068-1084)

Inside _summarize_groq:

for attempt in range(1, self.groq_max_retries + 1):

 try:

 response = self._groq_client.chat.completions.create(

 model=self.groq_model,

 messages=messages,

 max_tokens=300,

 temperature=0.2,

)

 generated = response.choices[0].message.content.strip()

 if generated:

 return generated

 except Exception as exc:

 last_error = exc

 msg = str(exc).lower()

 # Context-length errors: retry with smaller input

 if "context" in msg and ("length" in msg or "window" in msg or "token" in msg):

 break

 # Rate-limit: wait and retry

 if "rate" in msg and attempt < self.groq_max_retries:

 time.sleep(min(2.0 * attempt, 8.0))

 continue

 if self.require_llm:

 raise RuntimeError(f"Groq summarization failed: {exc}") from exc

```

```

 logging.warning("Groq summarization failed, using fallback: %s", exc)

 return self._fallback_summary(text)
 ...

Code: Ollama Transient Error Handling
```python
# pipeline.py (lines 838-858)
last_error: Optional[Exception] = None

for attempt in range(1, self.ollama_max_retries + 1):

    try:

        return self._ollama_post_json("/api/generate", payload)

    except Exception as exc:

        last_error = exc

        msg = str(exc).lower()

        transient = (

            "http error (500)" in msg

            or "http error (502)" in msg

            or "http error (503)" in msg

            or "http error (504)" in msg

            or "timed out" in msg

            or "temporary" in msg

            or "connection reset" in msg

        )

        if transient and attempt < self.ollama_max_retries:

            time.sleep(min(1.5 * attempt, 5.0))

            continue

        raise RuntimeError(f'Ollama generate failed: {exc}') from exc

```

...

Code: File and Input Validation

```python

# pipeline.py (lines 76-77) - File existence check

def parse\_document(self, pdf\_path: str, include\_media: bool = True) -> Dict:

if not os.path.exists(pdf\_path):

raise FileNotFoundError(pdf\_path)

# pipeline.py (lines 714-715) - Empty text guard

def summarize(self, text: str, context: str = "") -> str:

if not text.strip():

return "No content found for summarization."

# app.py (lines 185-206) - Upload parsing with error isolation

def parse\_uploaded\_files(uploaded\_files):

papers = st.session\_state.get("papers", {})

changed = False

for uploaded in uploaded\_files:

if uploaded.name in papers:

continue

with tempfile.NamedTemporaryFile(delete=False, suffix=".pdf") as tmp:

tmp.write(uploaded.getvalue())

pdf\_path = tmp.name

try:

data = services["extractor"].parse\_document(pdf\_path, include\_media=True)

papers[uploaded.name] = data

changed = True

except Exception as exc:

```

 logging.error("Parsing failed for %s: %s", uploaded.name, exc, exc_info=True)

 st.error(f"Unable to process **{uploaded.name}**. Skipping.")

 if changed:

 st.session_state["papers"] = papers

 return papers

...

```

## A.16 Project Directory Structure

```

Research-Paper-Summarizer/
|
|— app.py
|— pipeline.py
|— research_experiment_framework.py
|— run_research_experiments.py
|— requirements.txt
|— data/
|— outputs/
|— models/
|— static/

```

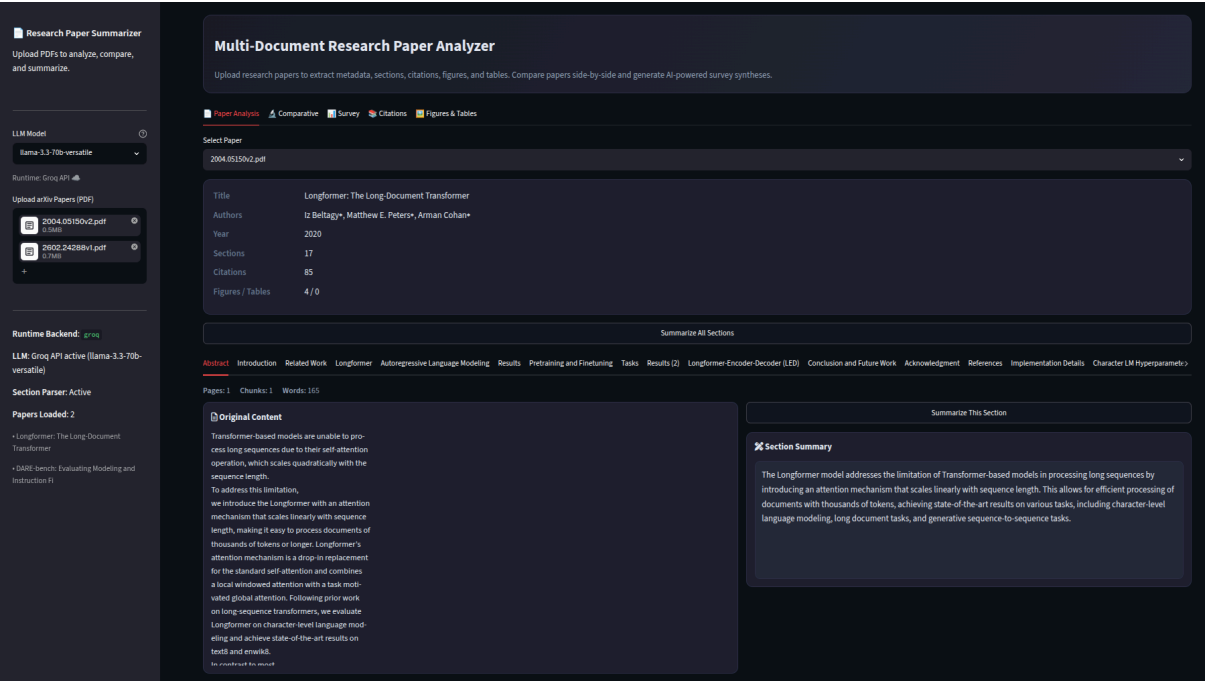
## A.17 Summary

The appended source code demonstrates the complete implementation of the Smart Research Paper Summarizer. The modular architecture enables efficient processing of research papers and supports advanced functionalities such as AI-powered summarization, comparative analysis, survey synthesis, citation extraction, and visual content analysis. The source code serves as evidence of implementation and may be referred to for future development and enhancement of the proposed system.

## APPENDIX B

### GRAPHICAL USER INTERFACE OUTPUTS

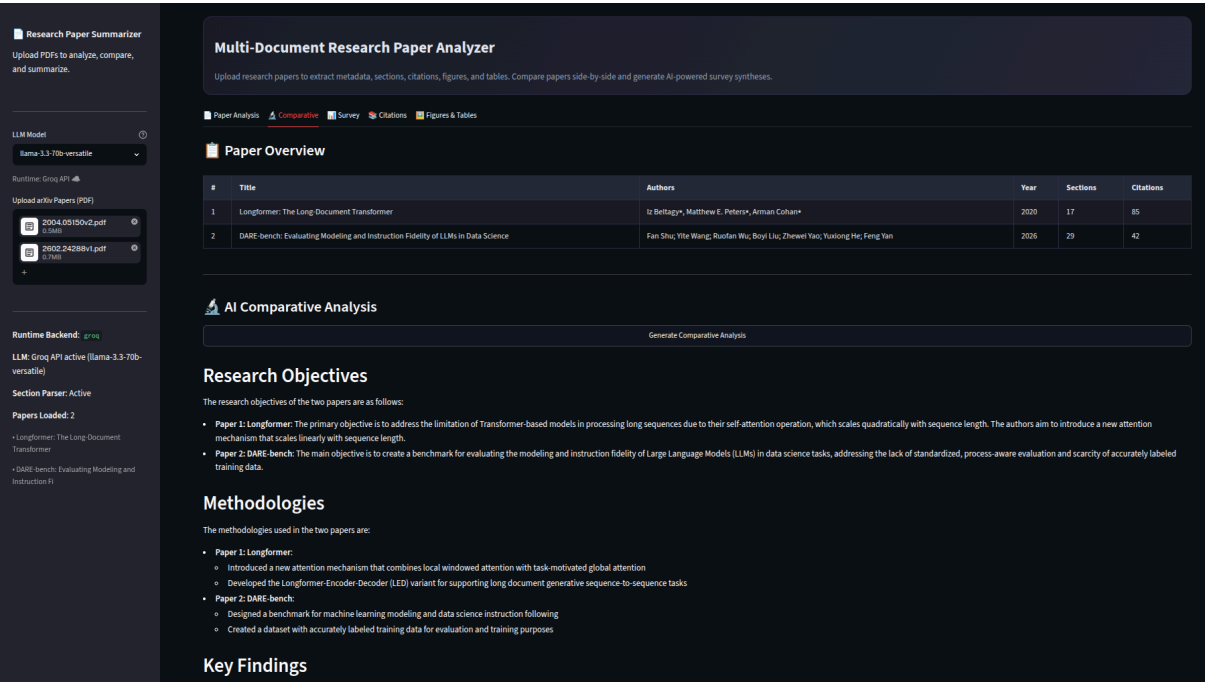
#### Figure B.1 Paper Analysis Dashboard



Description:

Displays metadata extraction, section-wise summaries, key findings, and research gaps generated from uploaded research papers.

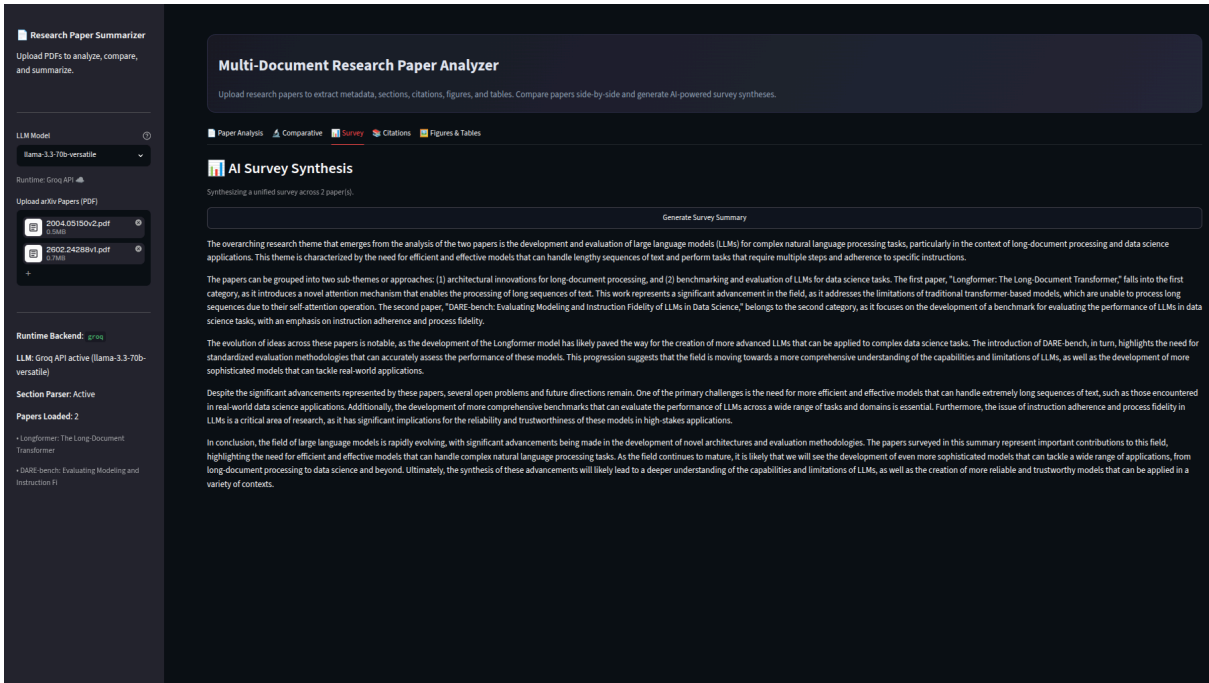
Figure B.2 Comparative Analysis Dashboard



Description:

Compares multiple research papers and generates AI-powered insights highlighting similarities, differences, methodologies, and contributions.

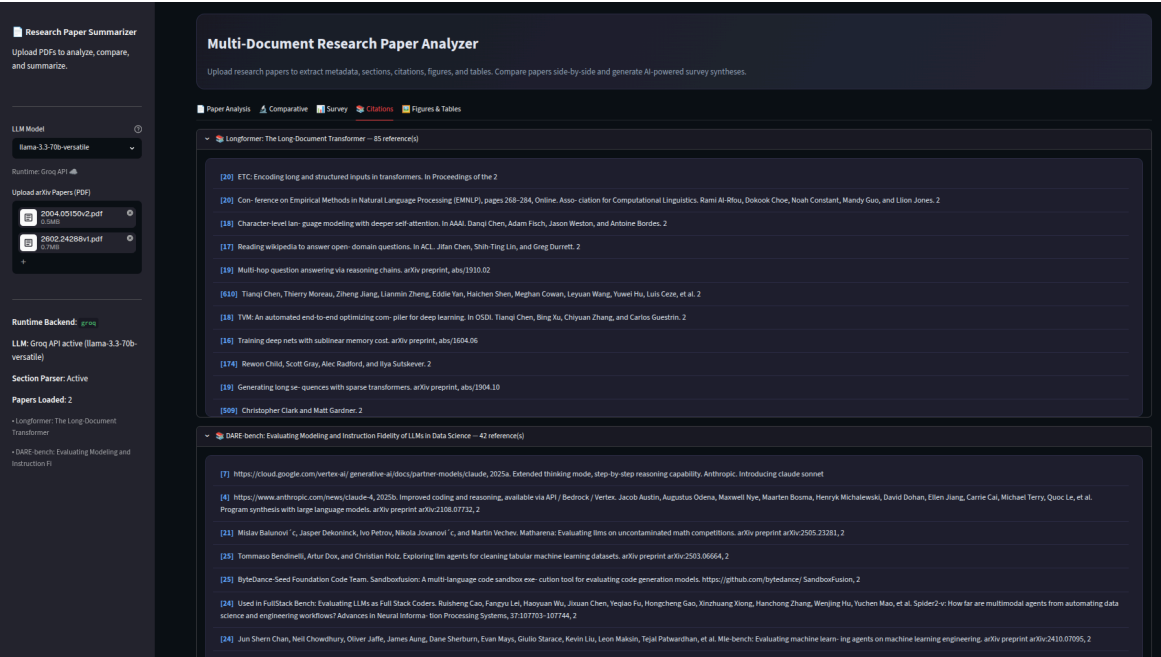
Figure B.3 Survey Synthesis Dashboard



Description:

Generates a consolidated survey report from multiple research papers and identifies common themes, trends, challenges, and future research opportunities.

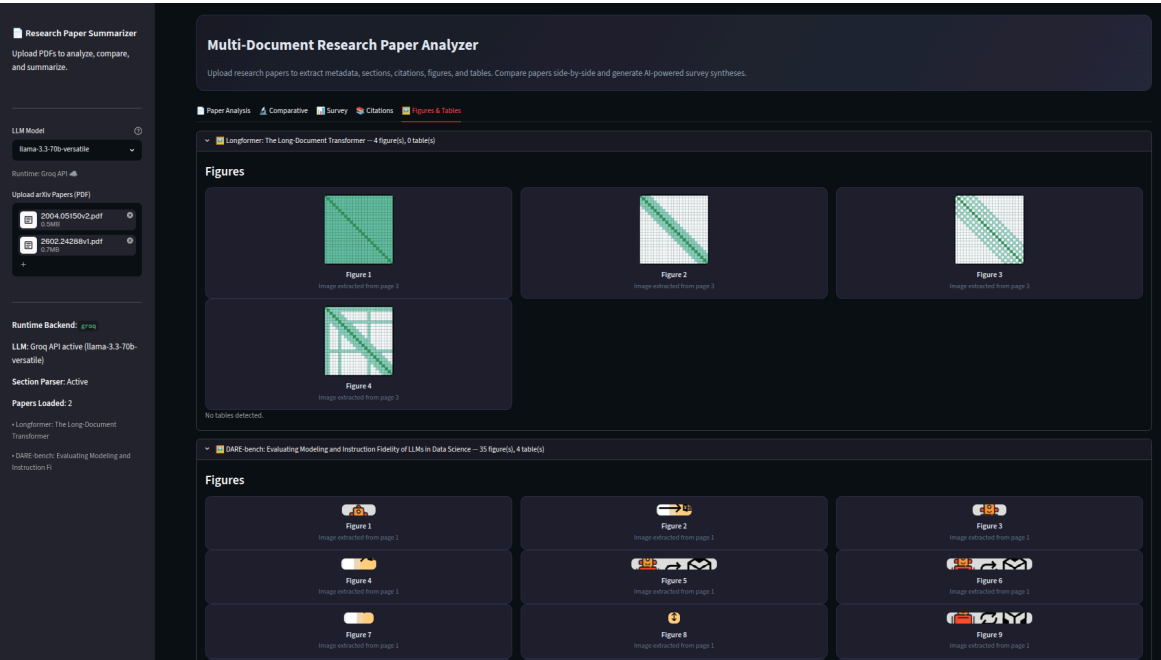
Figure B.4 Citation Analysis Dashboard



Description:

Extracts and analyzes references and citations from uploaded research papers for literature review support.

## Figure B.5 Figures and Tables Analysis Dashboard



Description:

Displays extracted figures, tables, charts, and captions from research papers to facilitate visual content analysis.

## **APPENDIX C**

### **SAMPLE OUTPUT GENERATED BY THE SYSTEM**

#### **Input Paper**

Longformer: The Long-Document Transformer

#### **Generated Summary**

##### **Research Objective**

To improve transformer performance on long documents while reducing computational complexity.

##### **Methodology**

Introduced a sparse attention mechanism combining local window attention and global attention.

##### **Key Findings**

Achieved superior performance on long document classification and question-answering benchmarks.

##### **Contributions**

- Proposed Longformer architecture.
- Reduced quadratic attention complexity.
- Improved long-context processing.

##### **Future Scope**

Integration with graph-based reasoning and multilingual document processing.